

Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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Introduction

Cluster randomized trials (CRTs) randomize intact groups, such as clinics, schools, or geographic areas, rather than individuals, and are the standard design when an intervention is delivered at the group level or when individual randomization would create unacceptable contamination between arms [1, 2]. A central assumption underlying standard CRT analysis is the no-interference component of the stable unit treatment value assumption (SUTVA): a cluster’s outcome must not depend on which other clusters were assigned to treatment [3]. This assumption, borrowed from individually randomized trial theory, can fail once clusters are spatially arranged, and so can the weaker assumption of partial interference, under which outcomes may depend on assignments within prespecified groups but not across them [4]. Tobler’s first law of geography, “everything is related to everything else, but near things are more related than distant things” [5], implies that an intervention delivered in one location can plausibly influence outcomes in geographically adjacent locations through shared populations, service networks, information diffusion, or environmental pathways. Documented examples of this kind of spatial spillover span public health and development economics: in a trial of insecticide-treated bed nets in northern Ghana, child mortality was lower in control villages close to treated villages than in

control villages farther away [6], and a school-based deworming trial in Kenya found health and attendance benefits among untreated children near treated schools [7]. When control clusters benefit from nearby treatment in this way, an unadjusted comparison of treated and control clusters may understate the direct effect, and how much contamination occurs can depend on the spatial arrangement of treated and control clusters, not only on the number of clusters allocated to each arm.

The design-stage gap in the spatial interference literature

The methodological literature addressing spatial interference in CRTs has concentrated on analysis-stage correction. Spatial autocorrelation statistics such as Moran’s I [8, 9] and Geary’s C [10] provide tools for detecting residual spatial dependence once outcomes have been observed, and spatial regression models, including spatial lag and spatial Durbin specifications [11], estimated by maximum likelihood [12] and implemented in modern software [13], allow an analyst to separate a direct treatment effect from a spillover effect after the fact. In their systematic review of spatial analysis methods for cluster randomized trials, Jarvis and colleagues placed trials that used spatial separation or buffers outside the primary review of statistical methods [14]. The literature therefore offers much more guidance on analyzing spatial effects than on how the choice of treatment assignment itself, made before any data are collected, might reduce the problem.

Design-stage tools do exist, but they were largely developed for a different purpose. Restricted randomization addresses the baseline imbalance that a limited number of clusters can produce [15]. Covariate-constrained randomization generates candidate assignments, retains those meeting a prespecified covariate-balance criterion, and randomizes among them [16]; Crisp and colleagues extended it to the selection of clusters from a larger eligible set and noted that their procedure also increases spatial dispersion among the selected clusters [17]. Their dispersion criterion concerns which clusters are selected rather than how the selected clusters are assigned to

arms, and none of these methods targets the spatial pattern of arm assignment under spillover: balancing a covariate across arms does not by itself determine whether treated and control clusters are spatially interspersed in a way that induces contamination. A more recent body of work addresses the problem studied in this chapter more directly. Randomized saturation designs assign different saturations, the fractions of units treated, to different groups and randomize treatment within each group, which allows direct and spillover effects to be jointly identified rather than merely controlled for [18]. A conventional “fried-egg” design omits each cluster’s outer area from analysis to limit contamination; McCann and colleagues proposed instead selecting whole clusters for inclusion or exclusion while preserving randomization [19]. Most recently, a group of theoretical papers published or posted between 2022 and 2025 has derived rate-optimal cluster-randomized designs under spatial interference [20], independent-set designs under network interference [21], designs that explicitly account for cross-cluster interference [22], and designs for spatio-temporally heterogeneous treatment effects [23]. This chapter complements this work with a simulation comparison of concrete assignment strategies that a trial team could implement, under heterogeneous baseline incidence and a range of spatial dependence and spillover conditions.

Extending a small-grid proof of concept

This chapter extends the proof-of-concept study of Chapter 2, which asked a narrower version of the same question: how does a priori treatment-assignment strategy affect estimation under spatial spillover and spatial dependence? That study compared simple random assignment against a single block-stratified design, on small, exhaustively enumerable grids of 8 to 12 spatial units arranged in 2×4 , 3×3 , and 3×4 configurations, and estimated the intervention effect with a spatial autoregressive (SAR) lag model containing a spillover indicator, with separate parameters for a baseline effect, the intervention effect, the spillover effect, and spatial dependence. Among its future considerations, Chapter 2 listed a broader range of grid dimensions and neighbor

structures beyond rook contiguity, and it proposed comparing simple random assignment and block-stratified assignment with buffer-zone (“fried-egg”) designs, which exclude from analysis the control clusters that border treated clusters. This chapter takes up a larger grid and queen contiguity, and it adds two extensions: several further assignment designs, none of which excludes clusters from analysis, and heterogeneous, rather than spatially uniform, baseline outcome risk. Chapter 2 suggested weighting assignment by cluster population density; some of the designs here use baseline outcome incidence instead. We scale from 8–12 clusters to a 100-cluster grid, from two designs to six conceptually distinct treatment assignment strategies (two further variants, which could not be distinguished from a retained design, are reported in Appendix A1), and from a single, uniform baseline risk surface to three distinct mechanisms of heterogeneous baseline incidence. This lets us ask which designs estimate the treatment effect well when baseline risk varies across space. Some of the designs use the incidence surface to assign treatment, and others ignore it.

Aims

This chapter has three aims. The first aim is to rank the six treatment assignment designs by estimation accuracy, measured by mean squared error (MSE), and by inferential validity, measured by 95% confidence interval coverage, separately within each of the three incidence-heterogeneity mechanisms and their five configurations, since a design that performs well under one mechanism may not generalize to another. The second aim is to test whether that ranking is stable across the spatial and spillover parameters that a real trial team cannot control or know in advance, including the magnitude of the true treatment effect itself, the strength of residual spatial dependence in the outcome, and the magnitude of the spillover effect. The third aim is to translate the resulting ranking into design guidance and describe how it will be applied in a planned sudden unexpected death (SUD) prevention pilot to be delivered through North Carolina’s community college network. The sections that follow describe the simulation methods,

report results for each aim, describe the planned SUD application, and discuss limitations and future work.

Methods

Spatial structure and outcome model

We simulated a 10×10 regular grid of 100 clusters under two definitions of adjacency. Under queen contiguity, two clusters are neighbors if they share an edge or a corner; under rook contiguity, they must share an edge. Queen contiguity is the primary analysis in this chapter. Rook contiguity is a sensitivity analysis and the case that continues Chapter 2, whose grids used rook neighbors. For each definition, the binary adjacency matrix was row-standardized to form the spatial weights matrix W , so that WZ gives, for each cluster, the proportion of its neighbors that are treated, and WY gives the average outcome among its neighbors. Row-standardization expresses each cluster's spillover exposure as an average over its own neighborhood rather than a raw sum, which would otherwise depend on how many neighbors the cluster has. Within a scenario, the same neighbor type defines W in the outcome model, in the Isolation Buffer design, and in the fitted model.

Cluster-level outcomes Y were generated from a spatial lag model with a spillover term,

$$Y = (I - \rho W)^{-1} [\tau Z + S(Z) + \beta X + \epsilon], \quad \epsilon \sim N(0, \sigma^2 I), \quad (1)$$

where $Z \in \{0, 1\}^{100}$ is the treatment assignment produced by the design under evaluation, τ is the direct treatment effect of interest, $S(Z)$ is the spillover term defined in Equation 2, ρ is the spatial autoregressive coefficient governing dependence in the outcome beyond that induced by treatment and spillover, X is the baseline incidence covariate with coefficient $\beta = 1$, and $\sigma = 1$. The spatial multiplier $(I - \rho W)^{-1}$ propagates each cluster's shocks through the network

121 of neighbors. We tested $\rho \in \{0, 0.01, 0.20, 0.50\}$. When $\rho = 0$ the multiplier is the identity and
 122 the model is a linear regression with independent errors, $Y = \tau Z + S(Z) + \beta X + \epsilon$, in which
 123 spatial structure enters only through the spillover term; larger values of ρ induce progressively
 124 stronger spatial clustering in the outcome.

125 We evaluated two spillover regimes,

$$S(Z) = \gamma WZ \quad (\text{both arms}), \quad S(Z) = \gamma WZ \circ (1 - Z) \quad (\text{control only}), \quad (2)$$

126 where $\circ$ denotes the elementwise product. Under the control-only regime, only untreated clusters
 127 respond to the treatment status of their neighbors, which represents classical contamination:
 128 an untreated cluster's outcome rises with the share of its neighbors that are treated. Under
 129 the both-arms regime, every cluster responds to its neighbors' treatment regardless of its
 130 own assignment, so neighboring treatment adds to a treated cluster's own effect. We tested
 131 four spillover magnitudes, $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, and five magnitudes of the direct effect,
 132 $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, with $\tau = 1$ as the primary scenario throughout.

133 **Heterogeneous baseline incidence**

134 Chapter 2 assumed a spatially uniform baseline outcome risk. Cluster-level health outcomes
 135 typically vary across space, and the designs studied here differ in whether and how they use that
 136 variation. We generated the baseline incidence covariate X , with values in $(0, 1)$, under three
 137 mechanisms. Under the independent (iid) mechanism, each X_i is drawn from a $\text{Uniform}(0, 1)$
 138 distribution, with no spatial structure. Under the spatially autocorrelated mechanism, $X =$
 139 $\Phi\{(I - \rho_X W_Q)^{-1}u\}$, where $u \sim N(0, I)$, Φ is the standard normal distribution function applied
 140 elementwise, W_Q is the queen weights matrix, and $\rho_X \in \{0.20, 0.50\}$. Under the Poisson
 141 mechanism, a spatially autocorrelated log-rate surface, $\log \lambda = \log(35/100,000) + (I - \rho_X W_Q)^{-1}u$
 142 with $\rho_X \in \{0.20, 0.50\}$, generates cluster death counts $C_i \sim \text{Poisson}(\lambda_i P)$, with an equal

143 population of $P = 100,000$ in every cluster, so that a cluster at the base rate expects 35 deaths.
 144 The observed rates C_i/P are then rank-normalized to $X_i = (r_i - 0.5)/N$, where r_i is the rank
 145 of cluster i 's rate among the $N = 100$ clusters and tied rates share their average rank. Ties
 146 remain at this population size: the Poisson surfaces used here have between 60 and 76 distinct
 147 values among the 100 clusters. The base rate of 35 per 100,000 was chosen to be representative
 148 of published SUD incidence: it lies between the SUD incidence rates that Mirzaei and colleagues
 149 estimated for US men (49.3 per 100,000) and women (21.7 per 100,000) aged 20–64, close to their
 150 average [24]. It is not the observed rate of the county data described in the Application section,
 151 which come from a different source, age range, period, and case definition. The incidence filters
 152 always use the queen weights W_Q , whichever neighbor type the scenario uses, so the incidence
 153 surfaces are identical under rook and queen contiguity. The iid mechanism and the two strengths
 154 of each of the other two mechanisms give five incidence configurations.

155 For each configuration we generated $K = 10$ incidence surfaces, $X_1, \dots, X_{10}$, each from its own
 156 seed. The same ten surfaces are used by every scenario in that configuration, across neighbor
 157 types, outcome-model parameters, designs, and values of τ ; the design draws and the outcome
 158 noise are shared more selectively, as described under Simulation design, resampling, and metrics.
 159 Each design is applied to the surface that generates the outcome: assignments for surface k
 160 are drawn from X_k , and the outcome and the fitted model for those assignments use X_k as
 161 well. This assumes that the incidence observed when a trial is planned is the incidence that
 162 drives its outcome (the matched-surface assumption). The target of each scenario is performance
 163 averaged over incidence surfaces; for example, $\text{MSE} = E_X E_{Z|X} E_\epsilon [(\hat{\tau} - \tau)^2]$, with X drawn from
 164 the configuration's generator, Z from the design given X , and $\epsilon \sim N(0, I)$. Bias, coverage, and
 165 power are defined in the same way.

Treatment assignment designs

We implemented eight treatment assignment designs and present six of them as the primary comparison set in this chapter. The other two, Saturation Quadrants and Balanced Halves, each share the mechanism of a retained design, and in the eight-design comparison at $\tau = 1$ neither could be distinguished from its counterpart under either neighbor type, by the Nemenyi and Holm-adjusted Wilcoxon signed-rank tests described under Simulation design, resampling, and metrics below, applied here to all eight designs. Saturation Quadrants assigns the saturations 20%, 40%, 60%, and 80% to the four quadrants in random order, without reference to incidence. Its mean squared error was 0.0900, against 0.0907 for Incidence-Guided Saturation Quadrants, under queen contiguity (Nemenyi $p = 0.98$; Holm-adjusted Wilcoxon signed-rank $p = 0.41$), and 0.0708 against 0.0717 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). We retain one representative of the saturation mechanism, Incidence-Guided Saturation Quadrants. Saturation Quadrants had the marginally lower mean under both neighbor types and the better average rank in the eight-design comparison (2.100 against 2.350 under queen contiguity and 2.281 against 2.306 under rook contiguity), so this choice does not rest on performance; that ordering the saturations by incidence makes no detectable difference is itself a result of this chapter. Balanced Halves divides clusters into two halves by incidence rank and treats 25 at random in each. Its mean squared error was 0.1299, against 0.1308 for Balanced Quartiles, under queen contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.88$), and 0.0842 against 0.0837 under rook contiguity (Nemenyi $p = 1.00$; Wilcoxon $p = 0.82$). The two designs differ only in how finely they stratify on incidence, and we retain the finer, four-stratum version. Balanced Halves had the marginally lower mean and the better average rank under queen contiguity (3.312 against 3.356), and Balanced Quartiles had both under rook contiguity (2.919 against 3.038), so this choice does not rest on performance either. Both excluded designs, and the full eight-design comparison, are reported in Appendix [A1](#).

Table 1 describes the six retained designs, and Figure 1 shows one assignment from each on a single illustrative incidence surface. Checkerboard is the strict block-stratified allocation of Chapter 2, in which no two rook neighbors share an arm. Chapter 2’s Block Stratified Sampling design also admitted relaxed allocations on its 3×3 grid, where the arms cannot be balanced; on its 2×4 and 3×4 grids, as on the 10×10 grid here, the strict allocations are the two checkerboard patterns. We use the two names interchangeably for this allocation and use Checkerboard in the rest of this chapter. Checkerboard is the only deterministic design: it gives the same assignment in every draw, although each of its fits still receives its own outcome noise. Every other design was drawn independently 25 times for each incidence surface. Designs that rank clusters by incidence break ties at random, afresh for each draw. High Incidence Focus is random only through that tie-breaking; under the iid and spatial mechanisms incidence values are distinct, so it gives the same assignment in all 25 draws on a surface, and its assignment changes only from one surface to the next. Isolation Buffer builds its assignment from the neighbor type of the scenario, and it treats 38 clusters on average under rook contiguity and 21 under queen contiguity. Every other design treats exactly 50.

Estimation

We estimated τ with two maximum likelihood spatial lag models, both fit to every simulated outcome and both including an intercept. The primary, oracle model has the spatial lag ρWY and the covariates Z , $\text{Spill} = S(Z)$, and X_k , so it matches the outcome model in Equation 1, including the true regime-specific spillover covariate from Equation 2. The oracle model assumes that the analyst knows the form of the spillover mechanism and has the exact exposure term, so its results are a best case for each design; we return to this assumption as a limitation. The non-oracle model, a sensitivity analysis, has the covariates Z and X_k only and omits the spillover term. Both models are fit to the same outcomes.

We fit both models with our own implementation of the maximum likelihood estimator of

Table 1: The six treatment assignment designs compared in this chapter, ordered by mean squared error under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (Table 2). A cluster’s incidence rank is its rank by the baseline incidence surface X_k , with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment; for Isolation Buffer, which depends on the neighbor type, it is the mean over its assignments under each neighbor type.

Design	Assignment rule	Clusters treated	Deterministic
Incidence-Guided Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants. Quadrants are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Quartiles	Clusters are divided by incidence rank into four strata of 25. Twelve clusters are treated at random in each stratum, and one more in each of two randomly chosen strata	50	No
Isolation Buffer	Clusters are treated one at a time, each chosen at random from those that are neither treated nor adjacent to a treated cluster, until none remain (a greedy random maximal independent set); no two treated clusters are neighbors	38 rook, 21 queen (mean)	No
High Incidence Focus	The 50 clusters with the highest incidence rank are treated	50	No
2×2 Blocking	The grid is divided into 25 non-overlapping 2×2 blocks; two of the four clusters in each block are treated at random	50	No
Checkerboard (strict Block Stratified Sampling allocation in Chapter 2)	Clusters whose grid coordinates (x, y) have an odd sum $x + y$ are treated, so treatment alternates from cluster to cluster; under rook contiguity every neighbor of a treated cluster is a control, and every neighbor of a control is treated	50	Yes

the spatial lag model [25], the same estimator that the `lagsarlm` function of the R package `spatialreg` computes with `method = "eigen"`. It maximizes the concentrated log-likelihood over ρ , computing the log-determinant from the eigenvalues of W [12], estimates the remaining coefficients by least squares given $\hat{\rho}$, and computes standard errors from the analytic information matrix. On 5,120 validation fits it agreed with `lagsarlm` to within 9.7×10^{-8} in $\hat{\tau}$ and to a relative difference of 1.1×10^{-8} in its standard error, at roughly 1% of the computing time; we also refit a 1% sample of the full simulation with `lagsarlm` itself as a check. Confidence intervals are 95% Wald intervals, $\hat{\tau} \pm 1.96 \widehat{SE}$.

Two combinations of design and neighbor type weaken or remove the information that separates

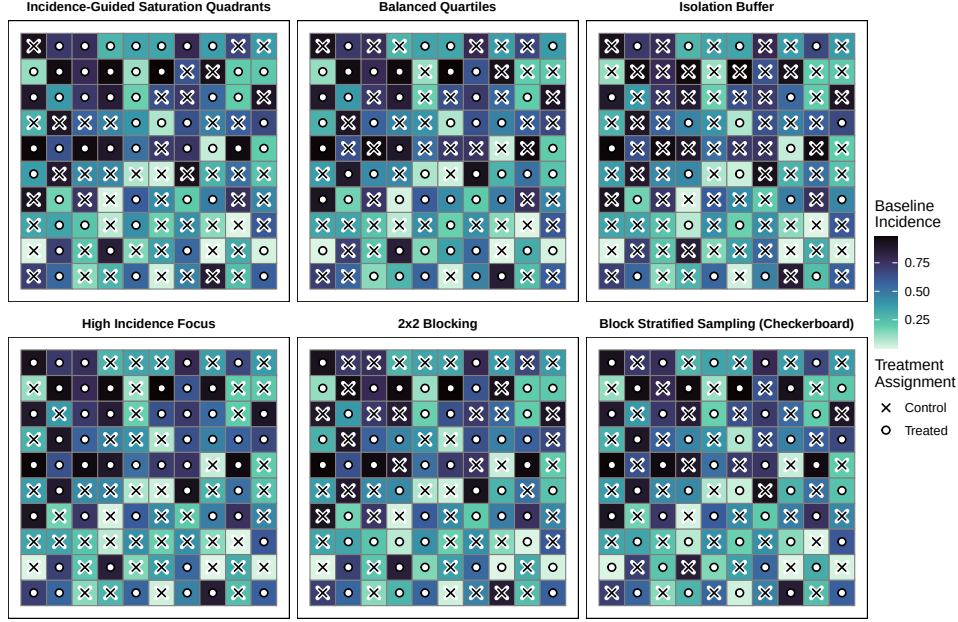


Figure 1: One sample assignment from each of the six designs on the 10 × 10 grid, with panels in the order of Table 1. All six assignments use a single illustrative incidence surface of 100 independent Uniform(0, 1) values generated with a fixed seed; it is not one of the simulation’s incidence surfaces. Tile fill shows baseline incidence (darker is higher), a ring marks a treated cluster, and an X marks a control. Isolation Buffer is shown under queen contiguity. The Checkerboard panel is labeled Block Stratified Sampling (Checkerboard), the Chapter 2 name for this allocation (see Treatment assignment designs).

τ from the spillover term. Under rook contiguity, Checkerboard gives $WZ = 1 - Z$ exactly,
 because every neighbor of a treated cluster is a control and every neighbor of a control is treated.
 Both spillover covariates then equal $\gamma(1 - Z)$, a linear combination of the intercept and Z , so τ
 is not identified: the fitting routine drops the aliased spillover column, and the coefficient on Z
 estimates $\tau - \gamma$. Chapter 2 noted the same consequence of exact collinearity between treatment
 and spillover exposure, $\hat{\beta} \approx \beta - \psi$ in its notation. The non-oracle model has no spillover column
 to drop, but its Checkerboard estimates under rook contiguity have the same defect. Under
 queen contiguity, each of the 64 interior clusters has four treated and four control neighbors, so
 $WZ = 1/2$ at every interior cluster. Under the control-only regime the spillover covariate then
 equals $(\gamma/2)(1 - Z)$ at those clusters, a linear combination of the intercept and Z , and only the
 36 clusters on the grid boundary carry information that separates it from the intercept and Z ;
 τ is identified, but imprecisely. Under the both-arms regime it is the constant $\gamma/2$ at interior

clusters, confounded with the intercept but not with Z . Isolation Buffer under rook contiguity can also, rarely, draw exactly the checkerboard pattern, a maximal independent set that equals one color class of the grid. This happened in 4 of its 40,000 rook draws and affected one fit in each of 20 scenarios, 4 at each value of τ . We flag every fit in which the fitting routine reports an aliased column and every scenario in which some draw has $\text{rank}([1, Z, WZ]) < 3$. Aliased estimates are kept in all summaries. In the exhibits, we flag Checkerboard under rook contiguity, where every fit is aliased; the Isolation Buffer scenarios, each with one aliased fit among its 250, are noted here and carry no flag in the exhibits.

Simulation design, resampling, and metrics

The full simulation crossed five values of τ , five incidence configurations, two neighbor types, four values of ρ , four values of γ , two spillover regimes, and eight designs, for $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12,800$ scenarios, of which the six designs reported in this chapter account for 9,600. We call a combination of incidence configuration, neighbor type, ρ , γ , and spillover regime a block; there are $5 \times 2 \times 4 \times 4 \times 2 = 320$ blocks, 160 for each neighbor type. At the primary scenario, $\tau = 1$, the six designs give 1,920 scenarios, and every block contains all six.

Within each scenario, we drew $J = 25$ assignments from each of the $K = 10$ incidence surfaces, for 250 fits per scenario, and gave every fit its own noise vector, so that fit (k, j) has outcome $Y_{kj} = (I - \rho W)^{-1}[\tau Z_{kj} + S(Z_{kj}) + \beta X_k + \epsilon_{kj}]$. Checkerboard's 250 fits share one assignment but have 250 distinct noise vectors. Given a surface, its 25 pairs of assignment and noise are independent, and the ten surfaces are independent of one another, so the surface is the primary sampling unit. Each estimator was fit 3.2 million times.

Every random draw was seeded from a hash of a key that names the draw: the configuration and surface index for an incidence surface; the block, surface, and design for a set of 25 assignments; and the block for the noise. Each draw therefore depends only on its key, and the results are identical whether scenarios are run sequentially, in parallel, or in any order. The keys also

give common random numbers. Within a block, all designs share the same noise vectors and incidence surfaces; the five values of τ reuse the same assignments and noise; and the oracle and non-oracle models are fit to the same outcomes. Assignments and noise are independent across blocks, but blocks within an incidence configuration share its ten surfaces.

For each scenario and estimator, we computed five metrics over the valid fits, those with a finite estimate and standard error: bias, the mean of $\hat{\tau} - \tau$; the standard deviation (SD) of $\hat{\tau}$; mean squared error, $\text{MSE} = \text{mean}\{(\hat{\tau} - \tau)^2\}$; coverage, the fraction of fits with $L \leq \tau \leq U$, where $L, U = \hat{\tau} \mp 1.96 \widehat{\text{SE}}$; and power, the fraction of fits with $L > 0$, a one-sided test of $\tau = 0$ at level 0.025. For both estimators, every one of the 250 fits in every scenario was valid, and the fitting routine raised no warning other than the aliasing flags described above.

Monte Carlo standard errors come from the surface-level metrics. For a metric m , we computed m_k on the 25 fits from surface k and set $\text{SE}(m) = \text{sd}(m_1, \dots, m_{10})/\sqrt{10}$, with intervals $m \pm t_{0.975, 9} \text{SE}(m)$. A difference between two designs is paired at the surface level: its standard error is that of $m_{d,k} - m_{d',k}$ over k . For a metric pooled over the five incidence configurations, we apply the same formula to the 50 configuration-by-surface means, each averaged over that surface's 32 blocks; these 50 means are independent of one another. Appendix A2 gives the parameter grid, the seed keys, and the exact formulas for these metrics and standard errors.

We compared designs within each neighbor type with a Friedman rank-sum test on scenario MSE [26], treating the 160 blocks as repeated measures across the six designs. At $\tau = 1$ we followed it with Nemenyi post-hoc comparisons [27, 28] and paired Wilcoxon signed-rank tests [29] with Holm's adjustment for multiple comparisons [30] for all 15 pairs of designs, and we repeated the Friedman test at each other value of τ . Blocks within an incidence configuration share their surfaces, so these tests treat dependent blocks as independent, which overstates the information about variation from surface to surface. For each pair of designs adjacent in the mean-MSE order, we therefore also ran a paired Wilcoxon signed-rank test over 50 independent

units, the 5 configurations times 10 surfaces, with each unit's MSE averaged over that surface's blocks, and a t_{49} interval for the mean difference. Because the values of τ share assignments and noise, Friedman tests at different values of τ are not independent confirmations of one another. We report results separately for each neighbor type, with queen contiguity primary, and no number is averaged over the two. Unless stated otherwise, every number in the Results is pooled over the five incidence configurations, and every table and figure caption states whether it pools the configurations or shows them separately. We report the ranking within each configuration separately and show the split wherever a conclusion changes.

Results

Design performance under queen contiguity

At the primary scenario, $\tau = 1$, under queen contiguity and pooled over the five incidence configurations, Incidence-Guided Saturation Quadrants had the lowest mean squared error (MSE = 0.091), followed by Balanced Quartiles (0.131), Isolation Buffer (0.160), High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087) (Table 2, Figure 2). Coverage of the 95% intervals was between 0.936 and 0.941 for all six designs (Figure 3). Power to reject $\tau = 0$ was 0.930 for Incidence-Guided Saturation Quadrants, 0.827 for Balanced Quartiles, 0.731 for Isolation Buffer, 0.642 for 2×2 Blocking, 0.610 for High Incidence Focus, and 0.547 for Checkerboard.

Incidence-Guided Saturation Quadrants also led block by block. Of the 160 queen blocks at $\tau = 1$, Incidence-Guided Saturation Quadrants had the lowest MSE in 95 (59%), Balanced Quartiles in 58 (36%), and Isolation Buffer in 7 (4%); no other design had the lowest MSE in any block.

Table 2: Performance of the six designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Monte Carlo standard errors of MSE, in parentheses, are computed from the 50 configuration-by-surface means (Methods); those of bias, coverage, and power are all below 0.007. Average rank is the mean of each design’s within-block MSE rank among the six designs over the 160 blocks (1 = lowest MSE). Clusters treated is the mean number of clusters treated per assignment. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged, and its coefficient on Z estimates $\tau - \gamma$.

Design	MSE (SE)	Bias	Coverage	Power	Average rank	Clusters treated
<i>Queen contiguity (primary)</i>						
Incidence-Guided	0.091 (0.0008)	0.032	0.937	0.930	1.425	50
Saturation Quadrants						
Balanced Quartiles	0.131 (0.0011)	0.017	0.936	0.827	2.138	50
Isolation Buffer	0.160 (0.0011)	0.003	0.939	0.731	3.694	21.4
High Incidence Focus	0.245 (0.0047)	0.031	0.938	0.610	4.763	50
2 $\times$ 2 Blocking	0.319 (0.0032)	0.010	0.939	0.642	4.244	50
Checkerboard	1.087 (0.0101)	0.001	0.941	0.547	4.737	50
<i>Rook contiguity (sensitivity)</i>						
Incidence-Guided	0.072 (0.0006)	0.018	0.942	0.963	1.488	50
Saturation Quadrants						
Balanced Quartiles	0.084 (0.0006)	0.008	0.940	0.917	1.775	50
Isolation Buffer	0.133 (0.0009)	0.003	0.939	0.806	3.381	38.2
High Incidence Focus	0.206 (0.0030)	0.018	0.939	0.645	4.969	50
2 $\times$ 2 Blocking	0.128 (0.0012)	0.004	0.938	0.818	3.400	50
Checkerboard (flagged)	0.480 (0.0015)	−0.652	0.150	0.414	5.987	50

Formal statistical comparison

The Friedman test rejected equal MSE ranks across the six designs under queen contiguity at $\tau = 1$ ($\chi^2 = 451.57$, $df = 5$, 160 blocks, $p < 2.2 \times 10^{-16}$). The average within-block ranks were 1.43 for Incidence-Guided Saturation Quadrants, 2.14 for Balanced Quartiles, 3.69 for Isolation Buffer, 4.24 for 2 $\times$ 2 Blocking, 4.74 for Checkerboard, and 4.76 for High Incidence Focus. Nemenyi comparisons (critical difference 0.596) separated 11 of the 15 pairs (Appendix A3 shows the critical-difference diagrams). Incidence-Guided Saturation Quadrants ranked significantly better than every other design, including Balanced Quartiles ($p = 0.0086$), and Balanced Quartiles ranked significantly better than the remaining four. The four pairs that Nemenyi comparisons did not separate all lay among the bottom four designs: 2 $\times$ 2 Blocking against Isolation Buffer

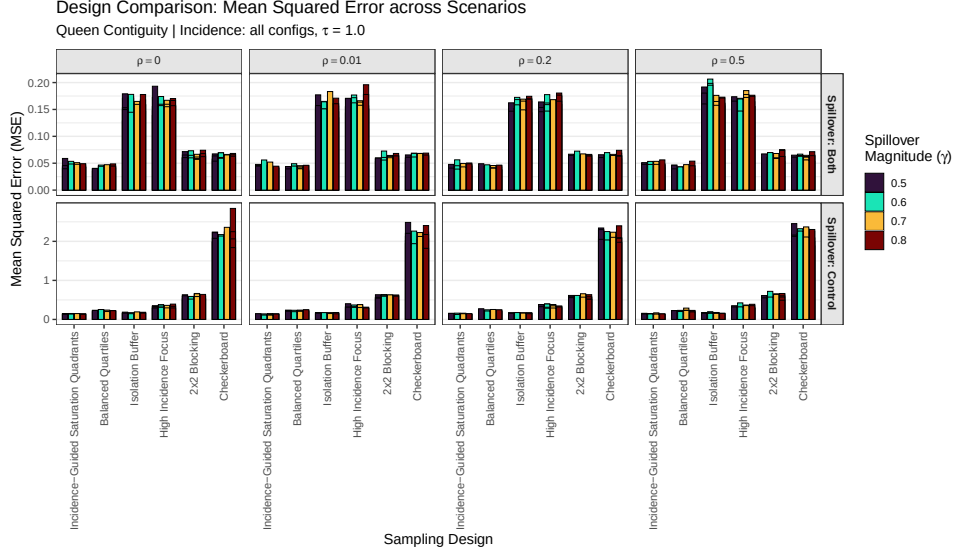


Figure 2: Mean squared error (MSE) of the oracle estimate of τ by design at $\tau = 1$, under queen contiguity. Columns of panels correspond to the four levels of outcome spatial dependence ρ , rows to the both-arms (top) and control-only (bottom) spillover regimes, and bar colors to the spillover magnitude γ . Each bar overlays the MSEs of the five incidence configurations for one combination of design, ρ , γ , and regime: the bar’s height is the largest of the five, and the horizontal lines within it mark the other four. Designs are ordered by pooled mean MSE (Table 2). The vertical scale is set separately for each row. Figure 9 in Appendix A4 shows rook contiguity.

($p = 0.090$), High Incidence Focus ($p = 0.130$), and Checkerboard ($p = 0.170$), and Checkerboard against High Incidence Focus ($p = 1.00$). Holm-adjusted Wilcoxon signed-rank tests separated all 15 pairs.

The bottom three designs were ordered differently by average rank and by mean MSE. By mean MSE the order was High Incidence Focus (0.245), 2×2 Blocking (0.319), and Checkerboard (1.087); by average rank it was 2×2 Blocking (4.24), Checkerboard (4.74), and High Incidence Focus (4.76). Both reversals have the same source: the MSEs of 2×2 Blocking and Checkerboard depended strongly on the spillover regime. Each had a lower MSE than High Incidence Focus in all 80 both-arms blocks (mean MSE 0.062 for each, against 0.162) and a higher MSE in all 80 control-only blocks (0.576 and 2.111, against 0.329; see the spillover-regime results below). Average rank counts each block once, so their wins under the both-arms regime offset their losses under the control-only regime; mean MSE weights those losses by their size.

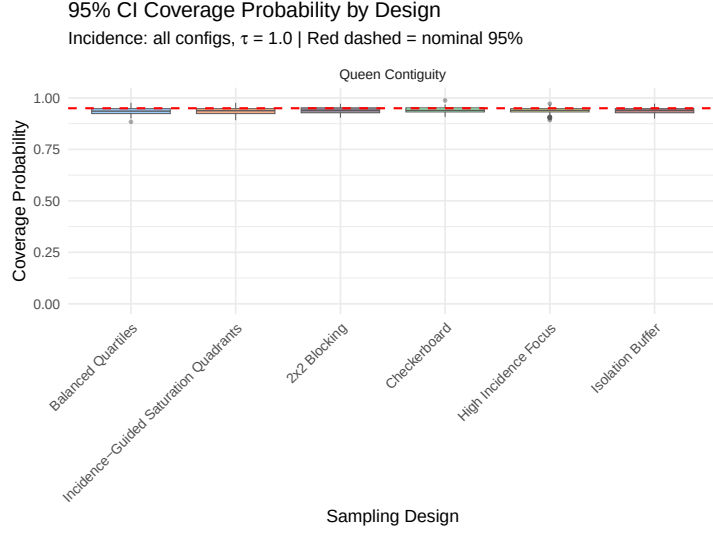


Figure 3: Coverage of the 95% confidence intervals by design at $\tau = 1$, under queen contiguity. Each box summarizes the coverage of the 160 scenarios of one design (5 incidence configurations $\times$ 4 values of $\rho \times$ 4 values of $\gamma \times$ 2 spillover regimes), each estimated from 250 fits. Designs are ordered by median coverage, so the order differs from the MSE order used elsewhere. The dashed red line marks the nominal 0.95 level. Figure 10 in Appendix A4 shows rook contiguity.

Because blocks within an incidence configuration share their ten surfaces, we also compared each pair of designs adjacent in the mean-MSE order over the 50 independent configuration-by-surface units. Every adjacent pair differed significantly. The mean paired differences in MSE, with t_{49} 95% intervals, were -0.0401 ($-0.0431, -0.0372$) for Incidence-Guided Saturation Quadrants against Balanced Quartiles, -0.0293 ($-0.0332, -0.0255$) for Balanced Quartiles against Isolation Buffer, -0.0853 ($-0.0954, -0.0752$) for Isolation Buffer against High Incidence Focus, -0.0738 ($-0.0858, -0.0617$) for High Incidence Focus against 2×2 Blocking, and -0.7677 ($-0.7893, -0.7460$) for 2×2 Blocking against Checkerboard, with Wilcoxon $p \leq 4.9 \times 10^{-9}$ in every case. The better design of each pair had the lower MSE in all 50 units, except High Incidence Focus against 2×2 Blocking, where it did so in 48 (96%). This test follows the mean-MSE order. It separated High Incidence Focus from 2×2 Blocking, which the Nemenyi comparison of within-block ranks did not, because each unit's MSE averages over both spillover regimes, and there the control-only losses of 2×2 Blocking outweigh its both-arms wins.

Design rankings by incidence configuration

Table 3 gives each design’s MSE within each of the five incidence configurations. Under queen contiguity, the mean-MSE order within every configuration was the same as the pooled order: Incidence-Guided Saturation Quadrants, Balanced Quartiles, Isolation Buffer, High Incidence Focus, 2×2 Blocking, and Checkerboard. The configuration changed each design’s MSE only slightly: in this order, each design’s highest MSE across the five configurations was lower than the next design’s lowest. Incidence-Guided Saturation Quadrants had the lowest MSE in 18 to 21 of the 32 blocks in each configuration. Coverage did not depend on the configuration either: within each configuration, every design’s coverage, averaged over its 32 scenarios, was between 0.931 and 0.943. The mechanism that generated the incidence surface therefore did not change the ranking. This includes High Incidence Focus, the design that uses the incidence ranks most directly: its MSE was no higher under the two Poisson configurations than under the iid and spatial configurations.

Table 3: Mean squared error by design within each of the five incidence configurations at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity). Each entry averages the 32 scenarios of one design, neighbor type, and configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. The spatially autocorrelated and Poisson configurations are labeled by the strength of dependence in the incidence surface, ρ_X . Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	iid	Spatial		Poisson	
		$\rho_X = 0.20$	$\rho_X = 0.50$	$\rho_X = 0.20$	$\rho_X = 0.50$
<i>Queen contiguity</i>					
Incidence-Guided Saturation Quadrants	0.093	0.087	0.092	0.090	0.092
Balanced Quartiles	0.130	0.130	0.131	0.128	0.135
Isolation Buffer	0.160	0.159	0.159	0.162	0.161
High Incidence Focus	0.245	0.267	0.245	0.232	0.238
2×2 Blocking	0.311	0.315	0.323	0.326	0.320
Checkerboard	1.091	1.068	1.086	1.112	1.077
<i>Rook contiguity</i>					
Incidence-Guided Saturation Quadrants	0.072	0.072	0.073	0.073	0.069
Balanced Quartiles	0.081	0.084	0.084	0.084	0.085
Isolation Buffer	0.131	0.131	0.132	0.136	0.134
High Incidence Focus	0.205	0.213	0.200	0.202	0.211
2×2 Blocking	0.129	0.125	0.129	0.130	0.129
Checkerboard (flagged)	0.480	0.482	0.474	0.481	0.482

Sensitivity to spatial dependence, spillover magnitude, and spillover regime

Under queen contiguity, neither the strength of spatial dependence in the outcome nor the spillover magnitude changed the ranking. Across $\rho \in \{0, 0.01, 0.20, 0.50\}$, no design's MSE changed by more than 0.02 (Table 4). Across $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$, the MSE of every design except Checkerboard also varied by less than 0.02, and Checkerboard's varied between 1.052 and 1.108 with no trend in γ (Table 5). Under rook contiguity, γ mattered for Checkerboard alone, for the reason given in the rook results below.

Table 4: Mean squared error by design across the four levels of outcome spatial dependence ρ , under queen contiguity at $\tau = 1$, pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\gamma \times$ 2 spillover regimes). Rows are ordered by pooled mean MSE under queen contiguity (Table 2).

Design	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Incidence-Guided Saturation Quadrants	0.090	0.087	0.093	0.093
Balanced Quartiles	0.129	0.129	0.132	0.133
Isolation Buffer	0.161	0.157	0.158	0.165
High Incidence Focus	0.243	0.242	0.249	0.248
2 $\times$ 2 Blocking	0.318	0.320	0.314	0.324
Checkerboard	1.095	1.077	1.084	1.092

Table 5: Mean squared error by design across the four levels of spillover magnitude γ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of $\rho \times$ 2 spillover regimes). Rows in both panels are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$
<i>Queen contiguity</i>				
Incidence-Guided Saturation Quadrants	0.092	0.089	0.092	0.089
Balanced Quartiles	0.132	0.128	0.132	0.131
Isolation Buffer	0.159	0.163	0.160	0.159
High Incidence Focus	0.244	0.249	0.248	0.241
2 $\times$ 2 Blocking	0.312	0.314	0.328	0.323
Checkerboard	1.108	1.052	1.086	1.101
<i>Rook contiguity</i>				
Incidence-Guided Saturation Quadrants	0.069	0.072	0.075	0.072
Balanced Quartiles	0.083	0.085	0.083	0.083
Isolation Buffer	0.131	0.133	0.134	0.133
High Incidence Focus	0.207	0.209	0.205	0.203
2 $\times$ 2 Blocking	0.129	0.129	0.127	0.128
Checkerboard (flagged)	0.295	0.403	0.538	0.684

The spillover regime was the one factor that changed the queen ranking (Table 6, Figure 2).

Under the both-arms regime, Balanced Quartiles had the lowest MSE (0.042), followed by Incidence-Guided Saturation Quadrants (0.046), 2×2 Blocking (0.062), Checkerboard (0.062), Isolation Buffer (0.161), and High Incidence Focus (0.162). Under the control-only regime, the order was Incidence-Guided Saturation Quadrants (0.135), Isolation Buffer (0.159), Balanced Quartiles (0.219), High Incidence Focus (0.329), 2×2 Blocking (0.576), and Checkerboard (2.111); at larger values of τ , Isolation Buffer matched or passed Incidence-Guided Saturation Quadrants under this regime (see Stability across treatment effect sizes). For Isolation Buffer, no treated cluster has a treated neighbor, so $WZ = 0$ at every treated cluster and $WZ \circ (1 - Z) = WZ$. The two regimes in Equation 2 therefore define the same outcome model, and so the same expected MSE; the observed values, 0.161 and 0.159, differ only by Monte Carlo error, because the two regimes use separate draws. Every other design had a higher MSE under the control-only regime, and the increase was largest for Checkerboard and 2×2 Blocking, which went from close to the best designs under the both-arms regime to the two worst under the control-only regime. Incidence-Guided Saturation Quadrants was first or second under both regimes, at this and every other value of τ , which matters because a trial team may not know in advance which regime applies.

Table 6: Mean squared error by design under each spillover regime at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 80 scenarios (5 incidence configurations $\times$ 4 values of $\rho \times$ 4 values of γ). Rows are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	Queen		Rook	
	Both arms	Control only	Both arms	Control only
Incidence-Guided Saturation Quadrants	0.046	0.135	0.045	0.098
Balanced Quartiles	0.042	0.219	0.042	0.125
Isolation Buffer	0.161	0.159	0.131	0.134
High Incidence Focus	0.162	0.329	0.164	0.249
2×2 Blocking	0.062	0.576	0.051	0.206
Checkerboard (flagged under rook)	0.062	2.111	0.477	0.483

The increase under the control-only regime was variance, not bias. The oracle model includes the true regime-specific spillover covariate, so it has no omitted-variable bias. At $\tau = 1$, in every

combination of design and regime under queen contiguity, the absolute bias was at most 0.048 and squared bias, averaged over scenarios, was at most 2.2% of MSE. Checkerboard showed the mechanism most clearly: under the control-only regime its spillover covariate is a linear combination of the intercept and Z at every interior cluster, so only the boundary clusters separate the spillover effect from the treatment effect (Methods). Checkerboard’s MSE rose from 0.062 under the both-arms regime to 2.111 under the control-only regime, while its bias stayed near zero (0.004) and its coverage stayed at 0.942: its intervals widened with its variance. Its pooled MSE of 1.087 was almost entirely variance: squared bias, averaged over scenarios, was 0.0045, less than 0.5% of MSE (Figure 4 shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). 2×2 Blocking showed the same pattern on a smaller scale, with MSE rising from 0.062 to 0.576 and a bias of 0.016 under the control-only regime.

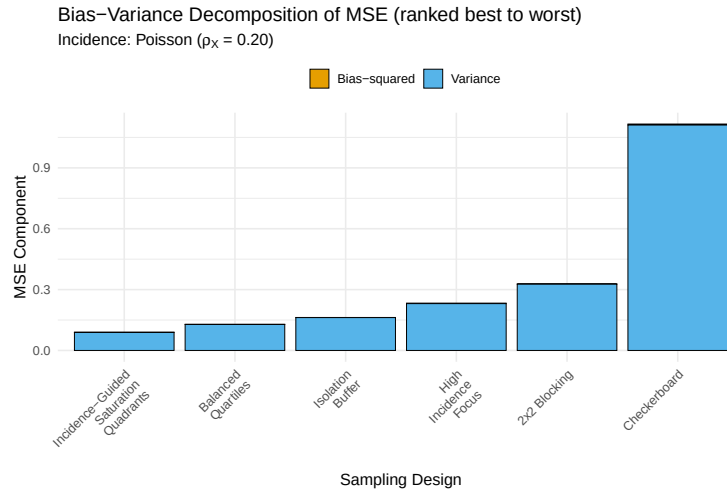


Figure 4: Decomposition of mean squared error into squared bias and variance by design at $\tau = 1$ for one incidence configuration, Poisson with $\rho_X = 0.20$, under queen contiguity. For each design, the two stacked components are the squared bias and the variance (SD^2) of $\hat{\tau}$, each averaged over the 32 scenarios of that configuration (4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes). Designs are ordered by pooled mean MSE (Table 2). Figure 11 in Appendix A4 shows rook contiguity.

Rook contiguity

Under rook contiguity at $\tau = 1$, pooled over the five incidence configurations, the mean-MSE order was Incidence-Guided Saturation Quadrants (0.072), Balanced Quartiles (0.084), 2×2

Blocking (0.128), Isolation Buffer (0.133), High Incidence Focus (0.206), and Checkerboard (0.480) (Table 2; Figure 9 in Appendix A4), and the same order held within each of the five incidence configurations (Table 3). Incidence-Guided Saturation Quadrants had the lowest MSE in 97 of the 160 rook blocks (61%) and Balanced Quartiles in 60 (38%). Coverage was between 0.938 and 0.942 for every design except Checkerboard, whose coverage was 0.150 (Figure 10 in Appendix A4); within single incidence configurations, the corresponding ranges were 0.935 to 0.946, and 0.14 to 0.15 for Checkerboard. The Friedman test rejected equal ranks ($\chi^2 = 703.76$, $df = 5$, 160 blocks, $p < 2.2 \times 10^{-16}$). Nemenyi comparisons separated 13 of 15 pairs; the two they did not separate were Incidence-Guided Saturation Quadrants against Balanced Quartiles ($p = 0.74$) and 2×2 Blocking against Isolation Buffer ($p = 1.00$). Holm-adjusted Wilcoxon tests separated every pair except 2×2 Blocking against Isolation Buffer. In the surface-level comparison, Incidence-Guided Saturation Quadrants had a lower MSE than Balanced Quartiles in 48 of the 50 units (mean difference -0.0121 ; -0.0139 , -0.0102 ; Wilcoxon $p = 9.9 \times 10^{-10}$). 2×2 Blocking and Isolation Buffer were close: 2×2 Blocking had the lower MSE in 36 of the 50 units (72%; mean difference -0.0048 ; -0.0078 , -0.0017 ; Wilcoxon $p = 0.0027$), but Isolation Buffer had the slightly better average rank (3.38 against 3.40), and the two designs traded places at some values of τ (see the results for τ below). Every other adjacent pair differed in all 50 units.

Checkerboard under rook contiguity is a different case: τ is not identified, and the coefficient on Z estimates $\tau - \gamma$ (Methods). The estimates behaved accordingly. At $\tau = 1$, Checkerboard's mean bias was -0.502 , -0.600 , -0.704 , and -0.802 at $\gamma = 0.5$, 0.6 , 0.7 , and 0.8 , while its mean standard deviation stayed between 0.20 and 0.21 ; squared bias, averaged over scenarios, accounted for 0.438 of its MSE of 0.480 (Figure 11 in Appendix A4 shows the decomposition for the Poisson configuration with $\rho_X = 0.20$). Its coverage fell from 0.316 at $\gamma = 0.5$ to 0.027 at $\gamma = 0.8$. Because $WZ = 1 - Z$, the two spillover covariates in Equation 2 coincide, $\gamma WZ \circ (1 - Z) = \gamma(1 - Z) = \gamma WZ$, so the two regimes define the same outcome model and the

same expected MSE; the observed values, 0.477 under the both-arms regime and 0.483 under the control-only regime, differ only by Monte Carlo error, because the two regimes use separate noise draws. Under queen contiguity, by contrast, Checkerboard’s coverage stayed between 0.939 and 0.943 at every level of ρ ; under rook contiguity it was between 0.134 and 0.161 (Table 7). Rook Checkerboard estimates were precise but centered on the wrong quantity, the opposite of its imprecise but valid queen estimates. We keep these estimates in every ranking and flag them.

Table 7: Checkerboard’s 95% confidence interval coverage across the four levels of outcome spatial dependence ρ at $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations $\times$ 4 values of γ $\times$ 2 spillover regimes). Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged. The nominal level is 0.95.

Neighbor type	$\rho = 0$	$\rho = 0.01$	$\rho = 0.20$	$\rho = 0.50$
Queen	0.941	0.943	0.940	0.939
Rook	0.161	0.156	0.148	0.134

Non-oracle sensitivity analysis

The non-oracle model, which omits the spillover term, was biased for every design. At $\tau = 1$, and setting aside Checkerboard under rook contiguity, its bias ranged from -0.12 to -0.40 and its coverage from 0.51 to 0.90 across designs and neighbor types. Under queen contiguity its bias ranged from -0.123 (Incidence-Guided Saturation Quadrants) to -0.244 (2×2 Blocking) and its coverage from 0.709 (2×2 Blocking) to 0.896 (High Incidence Focus); under rook contiguity the most biased design other than Checkerboard was Isolation Buffer (bias -0.400 , coverage 0.513). For Checkerboard under rook contiguity, the non-oracle estimates were identical to the oracle estimates (MSE 0.480, bias -0.652 , coverage 0.150), because once the aliased spillover column is dropped the two models are the same.

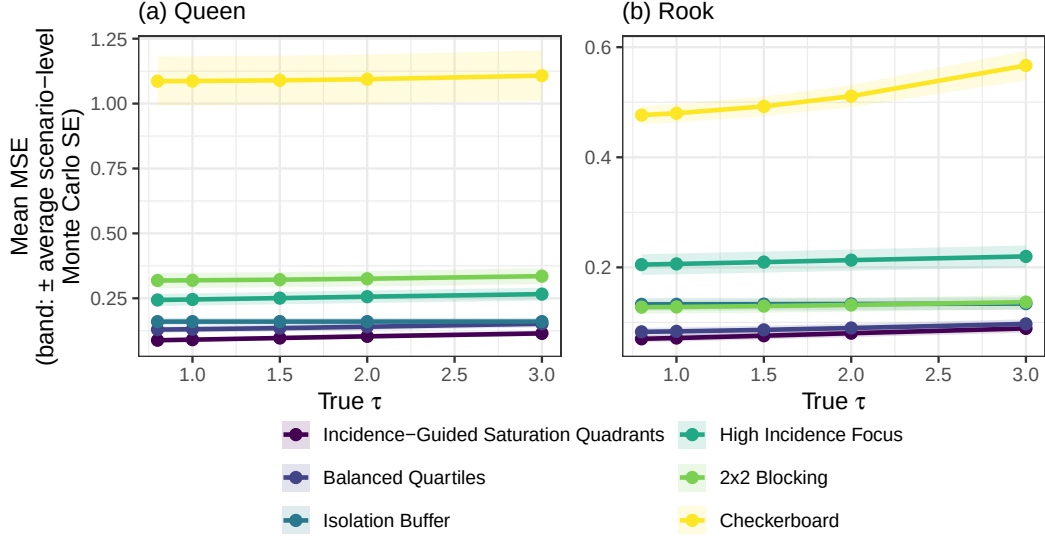
Omitting the spillover term also removes a column that can be nearly collinear with Z , and the non-oracle estimates were accordingly less variable for some designs. Under queen contiguity, the non-oracle MSE was lower than the oracle MSE for every design, most of all for Checkerboard (oracle 1.087, non-oracle 0.108) and 2×2 Blocking (oracle 0.319, non-oracle 0.121). Under rook

contiguity it was higher for every design except 2×2 Blocking (0.128 under both models) and Checkerboard (identical). These results do not change the recommendation. Incidence-Guided Saturation Quadrants also had the lowest non-oracle MSE under both neighbor types (0.087 under queen contiguity and 0.086 under rook contiguity), and where a design’s non-oracle MSE was lower, the estimate was biased and its intervals under-covered, so the lower MSE does not support valid inference.

Stability across treatment effect sizes

Under queen contiguity, each of the two design orders was the same at every $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$: the order by average within-block rank did not change across these values, and neither did the order by mean MSE, although the two orders differ from each other in the bottom three designs (see Formal statistical comparison; Figure 5a shows mean MSE). The Friedman statistic was 455.49, 451.57, 429.01, 411.19, and 376.79 at these five values ($p < 2.2 \times 10^{-16}$ at each). Under rook contiguity the order was also stable (Figure 5b) except for the close pair of 2×2 Blocking and Isolation Buffer: by average rank, 2×2 Blocking was ahead only at $\tau = 0.8$, and by mean MSE, Isolation Buffer was ahead only at $\tau = 3$. The five values of τ share their assignments and noise, so these five comparisons are not independent confirmations of one another; they show that, under queen contiguity, the order did not change across the five values of τ we tested, given the same draws.

Within that stable order, the margins moved. Under queen contiguity, the average rank of Incidence-Guided Saturation Quadrants rose from 1.41 at $\tau = 0.8$ to 1.85 at $\tau = 3$, but Balanced Quartiles did not gain on it: Incidence-Guided Saturation Quadrants had the lower MSE of the two in 103, 102, 102, 102, and 101 of the 160 blocks at $\tau = 0.8, 1, 1.5, 2$, and 3, and the average rank of Balanced Quartiles stayed between 2.13 and 2.14. Isolation Buffer gained instead, and its average rank fell from 3.71 to 3.31. The MSE of Isolation Buffer stayed at 0.160 at every value of τ , while that of Incidence-Guided Saturation Quadrants rose from 0.088 to 0.115 and



Rook: τ not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept but flagged.

Figure 5: Mean squared error by design across the five values of the treatment effect, $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each point averages the 160 scenarios of one design at one value of τ , pooled over the five incidence configurations. Shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of MSE; they describe the precision of a single scenario’s MSE, not of the plotted average, whose standard error is smaller (Table 2). The shared legend lists the designs in order of pooled mean MSE under queen contiguity at $\tau = 1$. The vertical scales of the two panels differ. The five values of τ share their assignments and noise.

that of Balanced Quartiles from 0.130 to 0.152, so the gap between Incidence-Guided Saturation Quadrants and Balanced Quartiles stayed between 0.037 and 0.041. Isolation Buffer’s gain came under the control-only regime, where Incidence-Guided Saturation Quadrants had a lower MSE than Isolation Buffer in 75, 73, 56, 40, and 12 of the 80 blocks at the five values of τ . The mean control-only MSE of Incidence-Guided Saturation Quadrants rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$, while that of Isolation Buffer stayed between 0.159 and 0.160 (Appendix A5). At $\tau = 2$ the two were level (0.1596 for Isolation Buffer against 0.1604), and at $\tau = 3$ Isolation Buffer had the lower control-only MSE (0.160 against 0.182). Incidence-Guided Saturation Quadrants remained second under the control-only regime at those two values, second to Balanced Quartiles under the both-arms regime at every value of τ , and lowest in pooled MSE at every value of τ . Its worst case across blocks was the lowest of the six designs at $\tau \leq 1.5$, but at $\tau = 2$ and $\tau = 3$ that of Isolation Buffer was lower (0.207 against 0.210, and 0.208 against 0.238). The mean bias

of Incidence-Guided Saturation Quadrants also grew with τ , from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$, but squared bias, averaged over scenarios, stayed at or below 3.0% of its MSE, so the rise in its MSE was mostly variance.

Under rook contiguity, Incidence-Guided Saturation Quadrants kept the lowest control-only MSE at every value of τ , although its margin over Isolation Buffer narrowed, from 0.095 against 0.134 at $\tau = 0.8$ to 0.131 against 0.136 at $\tau = 3$, where it had the lower MSE in 47 of the 80 control-only blocks. Its worst case across rook blocks was the lowest of the six designs at every value of τ except $\tau = 3$, where that of Isolation Buffer was lower (0.173 against 0.174).

Except for Checkerboard under rook contiguity, coverage stayed close to 0.94 at every value of τ . Pooled over the five incidence configurations, every design's coverage was between 0.936 and 0.941 at every value of τ under queen contiguity, and every design except Checkerboard was between 0.937 and 0.942 at every value of τ under rook contiguity. The coverage of rook Checkerboard, for which τ is not identified, rose from 0.138 at $\tau = 0.8$ to 0.150, 0.198, 0.269, and 0.442 at $\tau = 1, 1.5, 2$, and 3 (Appendix A5). Appendix A3 shows coverage across τ for each design.

Robustness across blocks and the recommended design

Table 8 gives the distribution of MSE across the 160 queen blocks at $\tau = 1$. Incidence-Guided Saturation Quadrants had the lowest median (0.080) and the lowest worst-case MSE of any design (0.169). Isolation Buffer had the narrowest range (0.128 to 0.206), because its MSE does not depend on the spillover regime, but at a higher level. Balanced Quartiles had a low median (0.115) and a worst case of 0.290. The distributions of 2×2 Blocking and Checkerboard reflected their dependence on the spillover regime: both had low first quartiles (0.062 and 0.063) and high upper quartiles (0.580 and 2.093). Checkerboard had the highest MSE in all 80 control-only blocks; in the 80 both-arms blocks, the highest MSE belonged to High Incidence Focus in 42 and to Isolation Buffer in 38. Under rook contiguity, Incidence-Guided Saturation Quadrants

505 also had the lowest worst-case MSE (0.138).

Table 8: Distribution of block-level mean squared error by design across the 160 blocks under queen contiguity at $\tau = 1$, pooled over the five incidence configurations (32 blocks in each). At a fixed value of τ , each block contributes one scenario per design. Rows are sorted by median, which here coincides with the order of pooled mean MSE under queen contiguity (Table 2).

Design	Best	Q25	Median	Q75	Worst
Incidence-Guided Saturation Quadrants	0.037	0.045	0.080	0.135	0.169
Balanced Quartiles	0.034	0.042	0.115	0.221	0.290
Isolation Buffer	0.128	0.150	0.159	0.169	0.206
High Incidence Focus	0.139	0.160	0.234	0.322	0.424
2×2 Blocking	0.048	0.062	0.271	0.580	0.719
Checkerboard	0.051	0.063	0.899	2.093	2.843

506 We recommend Incidence-Guided Saturation Quadrants. Under both neighbor types, it had
507 the lowest mean MSE, pooled over the two spillover regimes, in every incidence configuration
508 and at every value of τ , and it was first or second under each spillover regime at every value
509 of τ . At $\tau = 1$, its worst case across blocks was the lowest of the six designs, and its intervals
510 had coverage of 0.937 under queen contiguity and 0.942 under rook contiguity. Under queen
511 contiguity, Isolation Buffer, whose MSE does not depend on the regime, matched it under the
512 control-only regime at $\tau = 2$ and passed it at $\tau = 3$, but was fifth of the six designs under
513 the both-arms regime at every value of τ (fourth under rook contiguity). Balanced Quartiles
514 also performed well. It was second overall under both neighbor types at every value of τ , and
515 it was better than Incidence-Guided Saturation Quadrants only under the both-arms regime.
516 At $\tau = 1$ that advantage was marginal (0.0421 against 0.0459 under queen contiguity, 0.0421
517 against 0.0451 under rook contiguity), and Incidence-Guided Saturation Quadrants had the
518 lower MSE in 22 of 80 queen blocks and 23 of 80 rook blocks. Under the control-only regime,
519 Incidence-Guided Saturation Quadrants had the lower MSE in all 80 queen blocks and 75 of 80
520 rook blocks, by a wider margin (0.135 against 0.219 under queen contiguity).

Application: A Planned Sudden Unexpected Death Prevention

Pilot in North Carolina

Background

SUD in adults is an incompletely understood cause of premature mortality. The SUDDEN (Sudden Unexpected Death in North Carolina) methodology, a validated algorithm for classifying deaths as sudden and unexpected from death certificate and medical record data, was developed and screened against expert physician panel review by Nanavati and colleagues [31]. Using this methodology, Mirzaei and colleagues estimated SUD incidence rates of 49.3 per 100,000 among US men and 21.7 per 100,000 among US women aged 20–64, extrapolated from Wake County, North Carolina (see Heterogeneous baseline incidence). They found that lifetime productivity loss from SUD exceeded that from any individual cancer, and that among natural causes of death only all cancers combined and heart disease accounted for more years of life lost than SUD [24]. Subsequent work has documented geographic and socioeconomic gradients in SUD incidence within North Carolina: Gan and colleagues found elevated SUD incidence in rural counties relative to urban counties [32], and Mounsey and colleagues found an inverse relationship between household income and SUD incidence in Wake County [33]. SUD is therefore a measurable outcome in North Carolina whose incidence varies spatially and socioeconomically, which makes it relevant to the incidence-informed designs evaluated in this chapter.

The planned pilot design

A pilot intervention is being planned in collaboration with North Carolina’s community college (CC) system, which provides the spatial and organizational unit for the design framework developed in this chapter. North Carolina’s 58 community colleges collectively serve all 100 counties in the state; each college has a primary campus in one county and, in many cases, satellite campuses in neighboring counties (for example, Durham Technical Community College

operates a satellite campus in Orange County in addition to its primary Durham County campus). For this application, each county is assigned to exactly one CC service area, so the 58 service areas partition the 100 counties. Northampton and Bertie counties also lie within Roanoke-Chowan Community College's broader service network; we assign Northampton County to Halifax Community College, Bertie County to Martin Community College, and Hertford County alone to Roanoke-Chowan Community College. This mapping is a convention of this study's design, not a statement about the colleges' service commitments. Figure 6 shows the resulting partition of the 100 counties into the 58 service areas. This network structure motivates the spillover term in this chapter's outcome model. Community colleges within a shared service network communicate and coordinate programmatic activity with one another, so an intervention delivered at one college plausibly influences outcomes in a neighboring college's service area even without any formal expansion of the intervention itself. The spillover term in Equation 2 approximates this mechanism: it passes spillover between adjacent clusters through the contiguity weights W , whereas coordination between colleges follows their programmatic ties, which need not coincide with shared service-area boundaries.

The planned pilot site is Lenoir Community College, whose service area comprises Lenoir, Greene, and Jones counties. The pilot cluster is that whole service area, with anticipated spillover to adjacent CC service areas. The pilot intervention involves workforce development and disaster-response-style training intended to teach CC students and staff to recognize signs of medical distress, connect at-risk individuals to care, and support timely follow-up. The teaching intervention under consideration is delivered county-wide by design, rather than at finer spatial resolution such as the census tract level, so the treatment unit for this application is the CC service area (equivalently, a group of one or more counties) rather than the census tract. Individual-level incidence data may still be collected at finer resolution to inform the incidence-aware design components. The eventual evaluation plan anticipates that the intervention's effect will be visible first in intermediate process measures, such as medication adherence and the timeliness of clinic

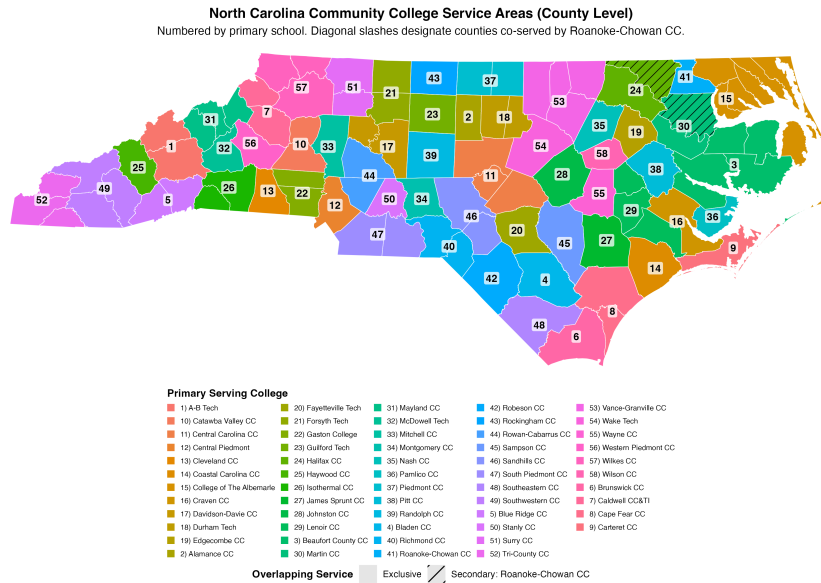


Figure 6: The assignment of North Carolina’s 100 counties to the 58 community college service areas used as clusters in the planned application. Each county is colored by its primary serving college, and each service area carries a single number, keyed to the legend. Hatching marks the two counties that also lie within Roanoke-Chowan Community College’s broader service network: Northampton County, assigned to Halifax Community College (24), and Bertie County, assigned to Martin Community College (30). Roanoke-Chowan Community College (41) serves Hertford County alone in this analysis. The planned pilot site, Lenoir Community College, is service area 29. The map shows county boundaries and the county-to-college assignment only; it does not depend on the SUD data.

visits following an at-risk event, since meaningful shifts in the SUD incidence rate itself are not expected to be detectable on the timescale of an initial pilot. This expected measurement lag between process outcomes and the ultimate incidence outcome modeled throughout this chapter is discussed further as a limitation below.

County-level SUD incidence for 2018–2021 was derived by Habib from North Carolina death certificates with the case definition of Habib [34], with one change. That definition excluded heart-failure deaths as presumed non-sudden, but some heart-failure patients were found to have died of adjudicated sudden cardiac death, so heart-failure deaths are included here, which makes the count larger than in that report (A. Habib, personal communication, September 25, 2026). Of the 412,514 deaths recorded statewide over that period, that derivation kept the 111,665 deaths among working-age adults (aged 18–64). It excluded deaths in hospitals and other institutions and deaths from expected causes (such as trauma, overdose, COVID-19, and

end-stage chronic disease, but not heart failure), and it applied a death-certificate algorithm validated against the SUDDEN physician-adjudicated cohort [32, 31]. The result is 23,523 sudden unexpected out-of-hospital deaths among working-age adults, pooled to the county level using death-certificate zip codes, with population-at-risk denominators from Surveillance, Epidemiology, and End Results (SEER) population estimates for ages 18–64 in the corresponding years. We have aggregated these county-level data to the 58 community college service-area clusters and computed cluster-level incidence for each year from 2018 to 2021 (see Data preparation and planned analysis). Applying the six designs to these incidence surfaces and comparing them is the next step and has not yet been done. Planned covariates for the eventual real-data application, once finalized, include county-level median income, food-desert status, physician access, and average driving distance to the nearest hospital.

Data preparation and planned analysis

The analysis has five steps, and the first three are complete. First, we aggregated the county-level SUD death counts and SEER populations aged 18–64 to the 58 CC service-area clusters, assigning each county to its single cluster. Second, we computed cluster-level SUD incidence for each year from 2018 to 2021, as the cluster’s deaths in that year divided by its population aged 18–64 in that year, per 100,000, and a pooled 2018–2021 rate per 100,000 person-years, as the cluster’s deaths summed over the four years divided by its population summed over the four years. Because heart-failure deaths are included, these counts exceed those reported by Habib [34]; the earlier counts behind that report, aggregated in the same way, reproduce its statewide death count, person-years, and yearly and overall rates, and in every year the 58 cluster totals sum to the statewide totals. Third, we built queen and rook contiguity weights for the 58 clusters, row-standardized as in the grid study; queen contiguity matches the primary analysis of this chapter. The weights use legal county boundaries rather than shoreline-clipped ones, so clusters whose counties meet across a river or sound are neighbors, because an education intervention

can plausibly reach nearby counties across water. Fourth, we will apply the six designs to the 58 clusters. The recommended design, Incidence-Guided Saturation Quadrants, is defined on the quadrants of a regular grid, so for the service-area geography its four quadrants are replaced by four regions: k-means clustering of the service areas' locations generates 250 candidate partitions, and the partition chosen is the one that best balances population and the numbers of clusters and counties across regions while keeping each region compact. The regions are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean, and within each region that share of its clusters is treated at random. The existing application code, which predates the revised simulation and is to be finalized, also adapts three other designs to the service-area geography: Checkerboard becomes a greedy two-coloring of the contiguity graph, in which each cluster, taken in decreasing order of its number of neighbors, is given the arm held by fewer of its already-assigned neighbors; 2×2 Blocking becomes k-means blocks of about four clusters, formed from the clusters' locations, incidence, and population, with half of each block, rounded, treated at random; and High Incidence Focus treats the 29 highest-incidence clusters, half of the 58. Fifth, we will compare the designs on the real cluster-level incidence surface, using the same simulation machinery as the grid-based comparison. The existing code for the comparison on the service-area geography predates the revised simulation described in Methods and has not yet been run on these data.

Application results (to be completed)

The results of this analysis will be reported here once it has been run.

Discussion

Summary and relevance

We compared six treatment assignment designs for a spatial cluster randomized trial on a grid of 100 clusters, under three mechanisms of heterogeneous baseline incidence, two spillover regimes, and a range of values for spatial dependence, spillover magnitude, and the treatment effect. Some findings held across the incidence configurations and the values of ρ , γ , and τ under queen contiguity. Others depended on the spillover regime, which a trial team may not know in advance.

Under queen contiguity, Incidence-Guided Saturation Quadrants had the lowest mean squared error (0.091) and Balanced Quartiles the second lowest (0.131). The order of the six designs was the same in each of the five incidence configurations and at every value of τ , and neither the strength of spatial dependence in the outcome nor the spillover magnitude changed it. The mechanism that generated the incidence surface changed each design's MSE only slightly. The same two designs also led under rook contiguity (0.072 and 0.084). Coverage of the 95% intervals was close to 0.94 for every design under both neighbor types, except Checkerboard under rook contiguity, so the designs differed in precision and coverage did not separate them.

Under queen contiguity, Checkerboard and 2×2 Blocking were close to the best designs when spillover reached both arms and were the two worst when it reached only control clusters: at $\tau = 1$, Checkerboard's MSE was 0.062 under the both-arms regime and 2.111 under the control-only regime, and that of 2×2 Blocking was 0.062 and 0.576. Balanced Quartiles also lost ground under the control-only regime (0.042 and 0.219), while Incidence-Guided Saturation Quadrants was first or second under both regimes at every value of τ . Its control-only MSE did rise with τ , and Isolation Buffer, whose MSE did not, matched it under that regime at $\tau = 2$ and passed it at $\tau = 3$. A design whose performance depends on the regime carries a risk that its average MSE does not show.

653 Checkerboard failed for different reasons under the two neighbor types. Under queen contiguity,
 654 its high MSE was variance, not bias: under the control-only regime, only the boundary clusters
 655 separate the spillover effect from the treatment effect (Methods). The estimates stayed centered
 656 on τ (bias 0.004 under the control-only regime at $\tau = 1$) and their intervals kept close to nominal
 657 coverage (0.942), but they were imprecise. Under rook contiguity, τ is not identified (Methods):
 658 the estimates were centered on $\tau - \gamma$, so rook Checkerboard's MSE grew with γ , and its coverage
 659 at $\tau = 1$ was 0.150. Checkerboard intersperses treated and control clusters as finely as the
 660 grid allows; both failures come from the dependence this pattern creates between Z and the
 661 spillover covariate, which is a linear combination of the intercept and Z at every cluster under
 662 rook contiguity and at every interior cluster under queen contiguity and the control-only regime.

663 We recommend Incidence-Guided Saturation Quadrants for the reasons given in the Results,
 664 chiefly that, under both neighbor types, it had the lowest pooled MSE and was first or second
 665 under each spillover regime at every value of τ , and that at $\tau = 1$ its worst case across blocks
 666 was the lowest of the six designs. Balanced Quartiles also worked well; its advantage under the
 667 both-arms regime was small, and its disadvantage under the control-only regime was large.

668 The saturation design did as well when its saturations were assigned without reference to
 669 incidence (Saturation Quadrants; see Treatment assignment designs). Both designs treat the
 670 four quadrants at saturations from 20% to 80%, but no design in this chapter removes that
 671 structure while keeping the rest fixed, so these results do not show that the saturation structure
 672 is the reason for their lower MSE.

673 These are design-stage results, and they complement analysis-stage modeling rather than replace
 674 it. Every design was analyzed with a spatial lag model that included the spillover term, and
 675 for every design whose τ was identified the estimates were close to unbiased: at $\tau = 1$, the
 676 absolute bias of each design under each regime was at most 0.048 under queen contiguity and
 677 0.027 under rook contiguity. Given that model, the choice of design determined how precisely τ

678 was estimated.

679 **Relation to Chapter 2**

680 Checkerboard is the strict allocation of the block-stratified design that Chapter 2 recommended
681 over simple random assignment on rook grids of 8 to 12 clusters. Chapter 2 used the term
682 “checkerboard paradox” for the amplified bias it observed for this design on its 3×4 grid when
683 spillover reached both arms and the outcome was spatially dependent. In its limitations, it also
684 noted that an allocation in which treatment and spillover exposure are exactly collinear gives
685 $\hat{\beta} \approx \beta - \psi$ in its notation. A rook checkerboard is such an allocation on a rectangular grid
686 of any size, and on 100 clusters its coefficient on Z estimated $\tau - \gamma$ in the same way. Under
687 queen contiguity, a neighbor structure that Chapter 2 listed as an extension, τ is identified, and
688 Checkerboard performed close to the best designs under the both-arms regime and worst under
689 the control-only regime. Chapter 2 compared the design only with simple random assignment;
690 against the wider set of designs studied here, Checkerboard had the highest mean MSE under
691 both neighbor types.

692 **Limitations**

693 These conclusions have several limitations. First, the primary estimator is an oracle. It includes
694 the exact spillover exposure term, so its results are a best case for each design. We fit a
695 non-oracle model that omits the spillover term as a sensitivity analysis. At $\tau = 1$, setting aside
696 Checkerboard under rook contiguity, it was biased for every design, with bias between -0.12
697 and -0.40 and coverage between 0.51 and 0.90 , and Incidence-Guided Saturation Quadrants
698 still had the lowest MSE under both neighbor types. The non-oracle model is only one departure
699 from the oracle. We did not study an analyst who includes a spillover term of the wrong form,
700 such as one built on the wrong neighbor definition or a distance decay.

701 Second, the designs and the fitted model used the incidence surface that generated the outcome

(the matched-surface assumption in Methods). In practice, planning incidence is estimated from an earlier period and measured with error. The designs that assign treatment by incidence rank, and the covariate adjustment, would then use a noisy surface, and we did not study how that changes the comparison.

Third, in the Poisson mechanism every cluster had the same population of 100,000. Real clusters, including the community college service areas in the Application, differ in population, so their observed rates differ in precision, and unequal populations may change how the designs that use incidence perform.

Fourth, all results come from a single regular 10×10 grid, with β and σ fixed at 1. We did not vary the grid's size or shape, and some results depend on its geometry, including the number of clusters Isolation Buffer treats and the 36 boundary clusters that identify τ for Checkerboard under queen contiguity and the control-only regime.

Fifth, the simulation shares random draws by design. Blocks within an incidence configuration share its ten incidence surfaces, so the block-level tests treat dependent blocks as independent, which is why we also compared adjacent designs over the 50 independent configuration-by-surface units. The five values of τ also share their draws, so the stability of the design order across τ is not an independent replication at each value (see Stability across treatment effect sizes).

Sixth, the simulation's base rate of 35 per 100,000 is not the rate of the county data described in the Application (see Heterogeneous baseline incidence). We kept it deliberately, because the simulation is a general design study, not a calibration to one data set. In the Poisson mechanism, incidence enters the designs and the fitted model only through the ranks of the observed rates, so the base rate affects the simulation only through the spatial pattern of those ranks and the number of tied rates.

Seventh, in the planned application, changes in SUD incidence are not expected to be detectable on the timescale of an initial pilot, so its evaluation will rely on intermediate process outcomes.

We evaluated the designs on a continuous cluster-level outcome generated by the model in Equation 1 and did not study how their performance carries over to a process outcome.

Future work

Four extensions follow from these limitations. First, a data-generating process in which the planning incidence is a noisy snapshot of a latent risk surface, and the outcome depends on the latent surface, would test the matched-surface assumption directly and show how the designs that assign treatment by incidence rank behave when those ranks are partly wrong. Second, unequal cluster populations in the Poisson mechanism would bring the simulation closer to the Application. Third, varying the size and shape of the grid, including irregular geographies such as the community college service areas, would show how much of the ranking depends on a regular 10×10 lattice. Fourth, and most immediately, applying the designs to the county-level SUD data described in the Application, now that those data are available, will show whether the recommendation holds under observed incidence.

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Appendix

A1. The eight-design comparison

Table 9 defines the two designs that we implemented but do not compare in the body of this chapter, in the format of Table 1, and Table 10 reports the comparison of all eight designs at the primary scenario, $\tau = 1$.

Table 9: The two designs implemented but not compared in the body of this chapter, in the format of Table 1. Saturation Quadrants shares its mechanism with Incidence-Guided Saturation Quadrants, and Balanced Halves with Balanced Quartiles (see Treatment assignment designs). A cluster's incidence rank is its rank by the baseline incidence surface X_k , with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment.

Design	Assignment rule	Clusters treated	Deterministic
Saturation Quadrants	The grid is divided into its four 5×5 coordinate quadrants, which are given saturations of 20%, 40%, 60%, and 80% in an order drawn at random for each assignment, without reference to incidence; within each quadrant, that share of its 25 clusters is treated at random	50	No
Balanced Halves	Clusters are divided by incidence rank into two strata of 50; 25 clusters are treated at random in each stratum	50	No

The argument for leaving Saturation Quadrants and Balanced Halves out of the body, with the Nemenyi and Wilcoxon results for each pair, is given under Treatment assignment designs in Methods; Table 10 supplies the rest of the eight-design comparison. Under both neighbor types, each excluded design is adjacent to its retained counterpart in mean MSE. The MSEs and coverages of the six retained designs are those of Table 2, given here to four and three decimal places.

A2. Simulation mechanics

This section gives the mechanical details behind Simulation design, resampling, and metrics (Methods): the parameter grid, the random-number streams, the Monte Carlo structure, and the exact formulas for the metrics and their Monte Carlo standard errors.

Table 10: Performance of all eight designs at the primary scenario, $\tau = 1$, under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations $\times$ 4 values of $\rho \times 4$ values of $\gamma \times 2$ spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Retained marks the six designs compared in the body of this chapter. Average rank is the mean of each design’s within-block MSE rank among the eight designs over the 160 blocks (1 = lowest MSE), so it differs from the six-design average rank in Table 2. Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.

Design	Retained	MSE	Coverage	Average rank
<i>Queen contiguity (primary)</i>				
Saturation Quadrants	No	0.0900	0.936	2.100
Incidence-Guided Saturation Quadrants	Yes	0.0907	0.937	2.350
Balanced Halves	No	0.1299	0.936	3.312
Balanced Quartiles	Yes	0.1308	0.936	3.356
Isolation Buffer	Yes	0.1602	0.939	5.138
High Incidence Focus	Yes	0.2455	0.938	6.763
2 $\times$ 2 Blocking	Yes	0.3192	0.939	6.244
Checkerboard	Yes	1.0869	0.941	6.737
<i>Rook contiguity (sensitivity)</i>				
Saturation Quadrants	No	0.0708	0.940	2.281
Incidence-Guided Saturation Quadrants	Yes	0.0717	0.942	2.306
Balanced Halves	No	0.0842	0.940	3.038
Balanced Quartiles	Yes	0.0837	0.940	2.919
Isolation Buffer	Yes	0.1329	0.939	5.225
High Incidence Focus	Yes	0.2062	0.939	6.969
2 $\times$ 2 Blocking	Yes	0.1281	0.938	5.275
Checkerboard (flagged)	Yes	0.4799	0.150	7.987

Parameter grid. Table 11 lists the parameters that were crossed to form the 12,800 scenarios and the settings that were held fixed.

Random-number streams. Every random draw in the simulation has its own seed, computed by hashing a key that names the draw. The key lists the draw’s type (incidence surface, assignments, or noise) and the parameters that identify it, with numeric values written to a fixed number of decimal places so that equal values always give the same key. The random-number generator and its methods for normal and discrete draws are also fixed explicitly, so the streams do not depend on software defaults. Table 12 lists the three kinds of draw, the parameters in their keys, and the scenarios that therefore share them. The consequences of this scheme, independence from the order of execution and common random numbers, are described in Methods.

Table 11: Parameter grid of the full simulation. The crossed parameters form $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 = 12,800$ scenarios, 9,600 of them for the six designs compared in the body of this chapter; the settings in the lower panel are the same in every scenario.

Parameter	Values	Levels
<i>Crossed</i>		
Treatment effect τ	0.8, 1.0 (primary), 1.5, 2.0, 3.0	5
Incidence configuration	iid; spatially autocorrelated, $\rho_X \in \{0.20, 0.50\}$; Poisson, $\rho_X \in \{0.20, 0.50\}$	5
Neighbor type	Queen (primary), rook (sensitivity)	2
Outcome spatial dependence ρ	0, 0.01, 0.20, 0.50	4
Spillover magnitude γ	0.5, 0.6, 0.7, 0.8	4
Spillover regime	Both arms, control only (Equation 2)	2
Design	The six designs of Table 1 and the two of Table 9	8
<i>Fixed</i>		
Grid	Regular 10×10 grid, $N = 100$ clusters	
Outcome model (Equation 1)	$\beta = 1$, $\sigma = 1$	
Poisson mechanism	Base rate 35 per 100,000; $P = 100,000$ per cluster	
Incidence surfaces per configuration	$K = 10$	
Assignments per surface	$J = 25$ (Checkerboard: one assignment, used in every draw)	
Fits per scenario and estimator	$KJ = 250$, each with its own noise vector	
Estimators	Oracle (primary) and non-oracle (sensitivity), both fit to every simulated outcome	

Table 12: Seed keys. Each key names the kind of draw and the parameters listed here; the incidence mechanism is iid, spatial, or Poisson, and the iid configuration, which has no ρ_X , uses 0 in its place. k indexes the incidence surface and d the design. A draw is shared by every scenario whose key parameters match its own.

Draw	Key parameters	Shared by
Incidence surface X_k	Incidence mechanism, ρ_X , k	Every block, design, and value of τ of the configuration
The $J = 25$ assignments of design d on surface k	Incidence mechanism, ρ_X , neighbor type, ρ , γ , spillover regime, k , d	All five values of τ and both estimators
The 250 noise vectors of a block	Incidence mechanism, ρ_X , neighbor type, ρ , γ , spillover regime	All eight designs, all five values of τ , and both estimators

Monte Carlo structure and checkpoints. Within a scenario, fit (k, j) , with $k = 1, \dots, 10$ and $j = 1, \dots, 25$, uses surface X_k , assignment Z_{kj} , and noise vector ϵ_{kj} , the (k, j) column of the block's 100×250 noise matrix (see Simulation design, resampling, and metrics). The unit of computation is a combination of incidence configuration, neighbor type, and ρ : 40 units, each covering 4 values of γ , 2 spillover regimes, 8 designs, 5 values of τ , 250 fits, and 2 estimators. The units ran in parallel, and the run stops if any unit fails. Each unit's results are saved together with a record of the parameter settings, the simulation code, and the software versions that

856 produced them, and saved results are reused only if all of these match the current run, so results
 857 from different code or settings are never combined.

858 **Scenario metrics.** For one scenario and estimator, let $f = 1, \dots, n$ index the valid fits,
 859 those with a finite estimate and standard error (here every fit was valid, so $n = 250$). Let
 860 $L_f, U_f = \hat{\tau}_f \mp z_{0.975} \widehat{\text{SE}}_f$, where $z_{0.975} = \Phi^{-1}(0.975) \approx 1.96$, and let $\bar{\tau} = n^{-1} \sum_f \hat{\tau}_f$. Then

$$\text{Bias} = \frac{1}{n} \sum_{f=1}^n (\hat{\tau}_f - \tau), \quad \text{SD} = \left\{ \frac{1}{n-1} \sum_{f=1}^n (\hat{\tau}_f - \bar{\tau})^2 \right\}^{1/2}, \quad \text{MSE} = \frac{1}{n} \sum_{f=1}^n (\hat{\tau}_f - \tau)^2,$$

861

$$\text{Coverage} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}\{L_f \leq \tau \leq U_f\}, \quad \text{Power} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}\{L_f > 0\},$$

862 so that $\text{MSE} = \text{Bias}^2 + (n-1) \text{SD}^2/n$.

863 **Surface-level standard errors.** For $m \in \{\text{Bias}, \text{MSE}, \text{Coverage}, \text{Power}\}$, let m_k be the same
 864 formula applied to the valid fits from surface k (here all 25). With $K = 10$ and $s(a_1, \dots, a_K) =$
 865 $\{(K-1)^{-1} \sum_k (a_k - \bar{a})^2\}^{1/2}$ the sample standard deviation,

$$\text{SE}(m) = \frac{s(m_1, \dots, m_K)}{\sqrt{K}}, \quad m \pm t_{0.975, K-1} \text{SE}(m).$$

866 Because every fit was valid, each surface contributes 25 fits and $m = K^{-1} \sum_k m_k$. For two
 867 designs d and d' in the same block at the same τ , the difference is paired at the surface level:
 868 with $\Delta_k = m_{d,k} - m_{d',k}$,

$$\text{SE}(m_d - m_{d'}) = \frac{s(\Delta_1, \dots, \Delta_K)}{\sqrt{K}},$$

869 with the same $t_{0.975, K-1}$ interval.

870 **Standard errors pooled over configurations.** For one design, neighbor type, and value
 871 of τ , let $\mathcal{B}_c$ be the 32 blocks of configuration c and $m_{b,k}$ the surface- k metric in block b . The

872 configuration-by-surface means and the pooled metric are

$$\bar{m}_{c,k} = \frac{1}{32} \sum_{b \in \mathcal{B}_c} m_{b,k}, \quad \bar{m} = \frac{1}{50} \sum_{c=1}^5 \sum_{k=1}^{10} \bar{m}_{c,k},$$

873 where $\bar{m}$ equals the mean of the metric over the 160 scenarios, and

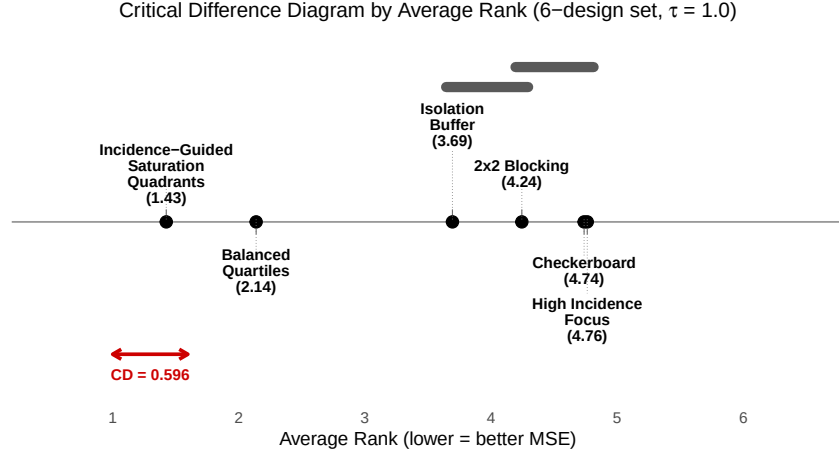
$$\text{SE}(\bar{m}) = \frac{s(\{\bar{m}_{c,k}\})}{\sqrt{50}},$$

874 with s taken over the 50 values $\bar{m}_{c,k}$. These are the standard errors in Table 2. For two designs d
875 and d' adjacent in the mean-MSE order (Formal statistical comparison), let $D_{c,k} = \bar{m}_{d,c,k} - \bar{m}_{d',c,k}$
876 for $m = \text{MSE}$. The interval for the mean difference is $\bar{D} \pm t_{0.975, 49} s(\{D_{c,k}\})/\sqrt{50}$, and the
877 Wilcoxon signed-rank test, with the normal approximation, is applied to the 50 values $D_{c,k}$.

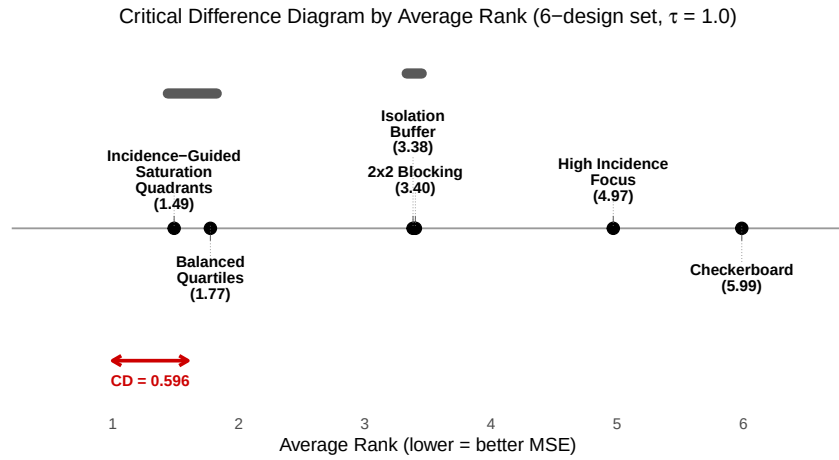
878 A3. Supplementary statistical comparisons

879 This section shows two figures behind results stated in the body of this chapter.

880 Figure 7 gives the Nemenyi critical-difference diagrams for the comparison of the six designs at
881 $\tau = 1$, whose results are reported under Formal statistical comparison (queen contiguity) and
882 Rook contiguity in the Results. Each design is placed at its average within-block MSE rank
883 among the six designs over the 160 blocks of its neighbor type. The red arrow shows the critical
884 difference, 0.596 at $\alpha = 0.05$, and a gray bar joins designs whose pairwise Nemenyi comparisons
885 were all not significant: under queen contiguity, Isolation Buffer with 2×2 Blocking, and 2×2
886 Blocking with Checkerboard and High Incidence Focus; under rook contiguity, Incidence-Guided
887 Saturation Quadrants with Balanced Quartiles, and Isolation Buffer with 2×2 Blocking. Where
888 two designs have nearly equal average ranks (Checkerboard and High Incidence Focus under
889 queen contiguity, Isolation Buffer and 2×2 Blocking under rook contiguity), the dotted leader
890 line to the outer label crosses the neighboring design's label; the positions and labels are correct.
891 Under rook contiguity, τ is not identified for Checkerboard; its estimates are kept and flagged.



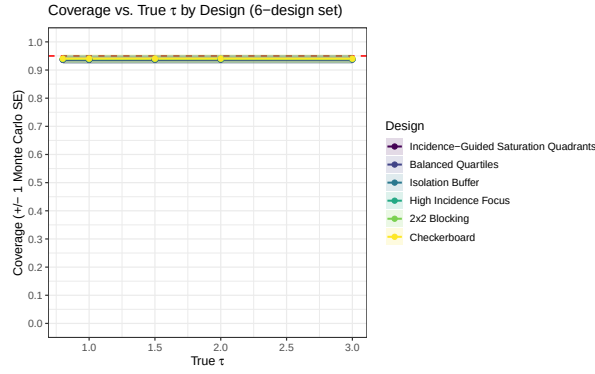
(a) Queen contiguity (primary)



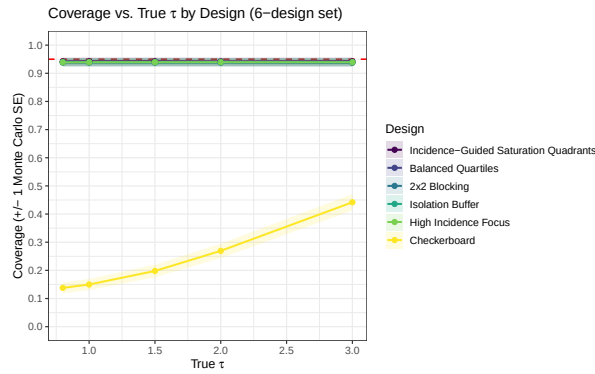
(b) Rook contiguity (sensitivity)

Figure 7: Nemenyi critical-difference diagrams for the six designs at $\tau = 1$, pooled over the five incidence configurations. Designs are placed at their average within-block MSE rank, given in parentheses (1 = lowest MSE); gray bars join designs that the Nemenyi comparisons did not separate.

Figure 8 gives the coverage of each design across the five values of τ , summarized under Stability across treatment effect sizes. The shaded bands show ± 1 times the average of the scenario-level Monte Carlo standard errors of coverage; they describe the precision of a single scenario's coverage, not of the plotted average. The legend lists the designs in order of pooled mean MSE at $\tau = 1$ for the neighbor type of the panel (Table 2). The vertical axis runs from 0 to 1, so lines close to 0.94 overlap: all six under queen contiguity, between 0.936 and 0.941, and all but Checkerboard under rook contiguity, between 0.937 and 0.942. The lowest line under rook contiguity, which the legend does not mark, is Checkerboard, for which τ is not identified; its



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 8: Coverage of the 95% confidence intervals by design across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$, pooled over the five incidence configurations. Each point averages the 160 scenarios of one design at one value of τ . The dashed red line marks the nominal 0.95 level.

estimates are kept and flagged. The five values of τ share their assignments and noise.

A4. Rook contiguity figures

This section gives the rook-contiguity versions of three figures shown for queen contiguity in the body of this chapter, as described under Rook contiguity in the Results. Each has the layout of its queen counterpart, but designs are ordered for rook contiguity, so the order differs from that of the queen figure. Under rook contiguity, τ is not identified for Checkerboard ($WZ = 1 - Z$); its estimates are kept and flagged. Figure 9 corresponds to Figure 2.

Figure 10 corresponds to Figure 3. Every design except Checkerboard has coverage close to the nominal level; Checkerboard's coverage, 0.150 pooled over the five incidence configurations (Table 2), is far below it.

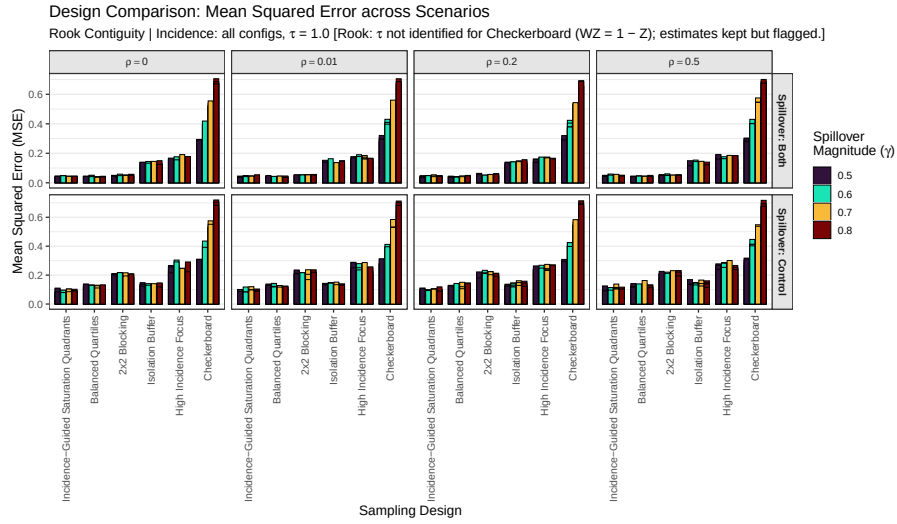


Figure 9: Mean squared error of the oracle estimate of τ by design at $\tau = 1$, under rook contiguity; layout as in Figure 2. Checkerboard is flagged (τ not identified).

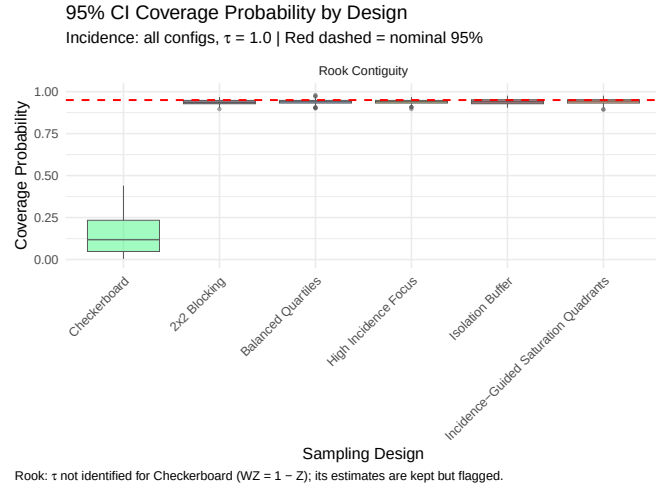


Figure 10: Coverage of the 95% confidence intervals by design at $\tau = 1$, under rook contiguity; layout as in Figure 3. Checkerboard is flagged (τ not identified).

Figure 11 corresponds to Figure 4. Nearly all of Checkerboard's MSE is squared bias, because its coefficient on Z estimates $\tau - \gamma$ (Methods).

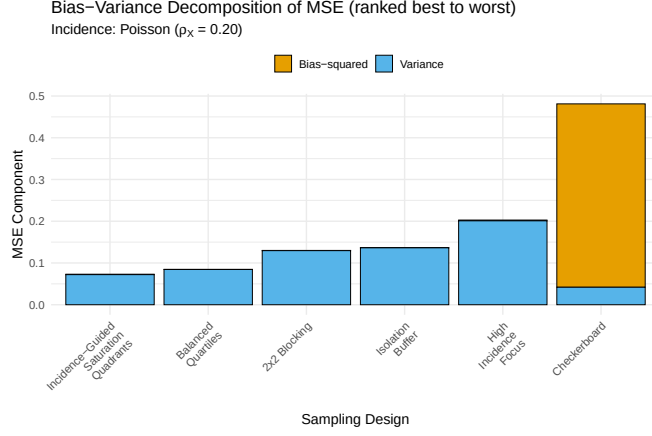


Figure 11: Squared bias and variance of $\hat{\tau}$ by design at $\tau = 1$ for the Poisson configuration with $\rho_X = 0.20$, under rook contiguity; layout as in Figure 4. Checkerboard is flagged (τ not identified).

A5. Technical notes

This section explains two results reported in the body of this chapter: the growth with τ of the bias of Incidence-Guided Saturation Quadrants, and the rise with τ of the coverage of Checkerboard under rook contiguity. The stored results keep neither $\hat{\rho}$ nor the standard error of each fit, so we re-ran every block for the six designs, both neighbor types, and all five values of τ , 2.4 million fits of the oracle model, recording $\hat{\rho}$ and the standard error, and the re-run reproduced the stored results in all 9,600 scenarios. Numbers below that involve $\hat{\rho}$, the standard error, or the terms of Equation 3 come from this re-run; the others come from the stored results.

Estimation error and its dependence on τ

For one fit (k, j) , write Y , Z , and ϵ for Y_{kj} , Z_{kj} , and ϵ_{kj} (Simulation design, resampling, and metrics). Let $V = [1, Z, S(Z), X_k]$ be the 100×4 covariate matrix of the oracle model, so that Equation 1 reads $(I - \rho W)Y = V\theta + \epsilon$ with $\theta = (0, \tau, 1, \beta)^\top$. Let $A = (I - \rho W)^{-1}$ and $V^+ = (V^\top V)^{-1}V^\top$, and for a vector v write $[V^+v]_Z$ for the coefficient on Z when v is regressed on the columns of V by least squares. Given $\hat{\rho}$, the estimated coefficients are $V^+(I - \hat{\rho}W)Y$

926 (Estimation). Because $(I - \hat{\rho}W)Y = V\theta + \epsilon + (\rho - \hat{\rho})WY$ and $WY = WA(V\theta + \epsilon)$,

$$\hat{\tau} - \tau = [V^+\epsilon]_Z + (\rho - \hat{\rho})h + \tau(\rho - \hat{\rho})g, \quad (3)$$

927 where

$$g = [V^+WAZ]_Z, \quad h = [V^+WA\{S(Z) + \beta X_k + \epsilon\}]_Z.$$

928 The first term has mean zero given Z and X_k . The last term carries a factor of τ . It is
 929 present exactly when $g \neq 0$, where g is the coefficient on Z when the lagged treatment $WAZ =$
 930 $WZ + \rho W^2Z + \rho^2 W^3Z + \dots$ is regressed on the model's own covariates, and its sign and size are
 931 set by the error in $\hat{\rho}$. The concentrated likelihood depends on Y only through the residual sum
 932 of squares of $(I - \tilde{\rho}W)Y$ on V , and the part of that residual that involves τ is $\tau(\rho - \tilde{\rho})$ times the
 933 residual of WAZ on V . If WAZ lies in the column space of V , that part vanishes for every $\tilde{\rho}$,
 934 so $\hat{\rho}$ is the same at every value of τ ; if in addition $g = 0$, then $\hat{\tau} - \tau$ is the same at every value of
 935 τ . Equation 3 held on every re-run fit, with a largest residual of 4.2×10^{-13} ; for Checkerboard
 936 under rook contiguity it held in the reduced form given below.

937 Two consequences of Equation 3 explain the results for τ reported in the body.

938 First, for designs whose treated clusters have treated neighbors, the last term gives a small
 939 positive bias that grows with τ . Under queen contiguity, pooled over the 160 blocks, the bias of
 940 Incidence-Guided Saturation Quadrants rose from 0.029 at $\tau = 0.8$ to 0.047 at $\tau = 3$ (Stability
 941 across treatment effect sizes), while that of Isolation Buffer stayed between 0.002 and 0.004.
 942 Almost all of the rise is the last term of Equation 3. In our fits, $\hat{\rho}$ was biased downward by
 943 about 0.06 on average (for Incidence-Guided Saturation Quadrants, by 0.071 at $\tau = 0.8$ and
 944 0.054 at $\tau = 3$). We observed this downward bias in these simulations; we did not derive it
 945 or check it against a published finite-sample result. With $\rho - \hat{\rho}$ positive on average, the last
 946 term is positive and grows with τ for any design with $g > 0$. For Incidence-Guided Saturation

947 Quadrants, g averaged 0.371 over the queen fits, and the last term contributed 0.021 of the
 948 bias at $\tau = 0.8$ and 0.043 at $\tau = 3$; the first term contributed 0.001 at both values, and the
 949 second 0.007 and 0.002. For Isolation Buffer, g averaged 0.020, and the last term contributed
 950 0.001 and 0.004. Most of the difference in g comes from the control-only regime. Under the
 951 both-arms regime with $\rho = 0$, the lagged treatment is $WZ = S(Z)/\gamma$, one of the covariates,
 952 so $g = 0$ for every design. Under the control-only regime, $WZ = S(Z)/\gamma + Z \circ WZ$, and at
 953 $\rho = 0$, g is close to the average share of treated neighbors among treated clusters. That share
 954 is 0.57 for Incidence-Guided Saturation Quadrants, whose treated clusters are concentrated in
 955 the high-saturation quadrants, and exactly 0 for Isolation Buffer, whose treated clusters have
 956 no treated neighbors. For Isolation Buffer, $Z \circ WZ = 0$, so the two regimes define the same
 957 spillover covariate and the same g , which is 0 at $\rho = 0$ and about 0.06 at $\rho = 0.5$; at $\rho = 0$, its $\hat{\rho}$
 958 and its estimation error were the same at every value of τ , up to the optimizer's tolerance.

959 The same term explains why the control-only MSE of Incidence-Guided Saturation Quadrants
 960 rose from 0.131 at $\tau = 0.8$ to 0.182 at $\tau = 3$ while that of Isolation Buffer did not. Under that
 961 regime, averaged over the 80 queen blocks, the variance of the last term of Equation 3 rose from
 962 0.008 to 0.061 for Incidence-Guided Saturation Quadrants, while the variance of the rest of the
 963 error changed little (0.119 and 0.115). The variance of the last term grew more slowly than τ^2
 964 because $\hat{\rho}$ became less variable as τ grew. For Isolation Buffer, the variance of the last term
 965 stayed below 0.0002. The rise in bias slowed at larger values of τ for a related reason: the part
 966 of the likelihood that involves τ sharpens it around the true ρ as τ grows, so the downward bias
 967 of $\hat{\rho}$ shrank.

968 Second, the decomposition explains why the coverage of Checkerboard under rook contiguity
 969 rose with τ . Under rook contiguity, Checkerboard's coverage rose from 0.138 at $\tau = 0.8$ to
 970 0.442 at $\tau = 3$ (Stability across treatment effect sizes), although its coefficient on Z estimates
 971 $\tau - \gamma$ at every value of τ . For this design $WZ = 1 - Z$, and because W is row-standardized,

972 $W^2Z = W(1 - Z) = Z$. The powers of W applied to Z therefore alternate between $1 - Z$
973 and Z , and $WAZ = \{(1 - Z) + \rho Z\}/(1 - \rho^2)$ lies in the span of the intercept and Z . The
974 derivation of Equation 3 applies with the aliased spillover column dropped, $V = [1, Z, X_k]$ and
975 $\theta = (\gamma, \tau - \gamma, \beta)^\top$, so that $\tau - \gamma$ takes the place of τ , and $g = -1/(1 + \rho)$. Two consequences
976 follow: $\hat{\rho}$ is the same at every value of τ , and

$$\hat{\tau} - \tau = -\gamma + \delta + (\tau - \gamma) \frac{\hat{\rho} - \rho}{1 + \rho},$$

977 where δ collects the first two terms of that decomposition and, like $\hat{\rho}$, does not depend on τ . The
978 error in $\hat{\rho}$ is therefore multiplied by $\tau - \gamma$, which grows with τ . The mean standard deviation
979 of $\hat{\tau}$ rose from 0.202 at $\tau = 0.8$ to 0.327 at $\tau = 3$, and the standard error tracked it, rising on
980 average from 0.199 to 0.322. With the bias held near $-\gamma$, wider intervals covered τ more often.
981 The rise was largest at $\rho = 0$ (coverage 0.145 to 0.541) and smallest at $\rho = 0.5$ (0.130 to 0.265),
982 consistent with the factor $1/(1 + \rho)$. The same term raised the MSE from 0.477 to 0.567. About
983 three quarters of that increase was variance, $(n - 1)SD^2/n$ averaged over scenarios (Appendix
984 A2), which rose from 0.041 to 0.109. The other quarter was squared bias, which rose from 0.436
985 to 0.458 because $\hat{\rho}$ was biased slightly downward, moving the mean bias from -0.650 to -0.667 .
986 The higher coverage at large τ reflects wider intervals around the wrong quantity, not a recovery
987 of τ .